

The School of Mathematics



THE UNIVERSITY
of EDINBURGH

Density-Aware Graph Geodesic Stress Testing on a Learned Macroeconomic Manifold

by

Adam Fowlie

Dissertation Presented for the Degree of
MSc in Computational Applied Mathematics

August 2026

Supervised by
Dr. Usama Nadeem & Dr. Etienne Hofstetter

Abstract

[abstract]

Acknowledgments

[acknowledgments]

Own Work Declaration

[own work declaration]

AI disclosure

[see guidance document]

Contents

1	Introduction	1
1.1	motivation	1
1.2	Research questions	2
1.3	contributions	2
1.4	Findings	2
2	Background	3
2.1	High-dimensional state vectors and the manifold hypothesis	3
2.2	Latent diffusions and spectral coordinates	3
2.3	Diffusion maps	3
2.4	Kernel scale and spectral diagnostics	6
2.5	Graph geometry and shortest paths	7
2.6	Density-aware graph geometry??	7
2.7	Hidden Markov models??	8
3	Data and Preprocessing	9
3.1	Data sources and sample period	9
3.2	Variable selection	9
3.3	Temporal alignment and missing observations	10
3.4	Transformations	10
3.5	Standardisation	10
3.6	Final analysis matrix	11
4	Numerical methods	12
4.1	Overview of the computational pipeline	12
4.2	Diffusion-map construction	12
4.3	Diffusion parameter selection	14
4.4	Graph construction and density estimation	16
4.5	Stress-path construction	17
4.5.1	Endpoint selection	17
4.5.2	Path definitions	18
4.6	Stress-path evaluation	19
4.7	Lifting paths into macroeconomic space	20
4.8	Nyström extension for out-of-sample embedding	22
4.9	Gaussian HMM specification and evaluation	24
4.10	Principal component benchmark	26
4.11	Robustness design	27
4.11.1	Sensitivity to core path parameters	27
4.11.2	Sensitivity to density estimation	28
4.11.3	HMM robustness	30
5	Numerical Validation on Toy Model	31
5.1	Purpose of the validation experiments	31
5.2	Benchmark manifolds	31
5.3	Bandwidth and spectral validation	31
5.4	Recovery of manifold geometry	31
6	Macroeconomic Manifold and Stress Paths	32
6.1	Empirical embedding results	32
6.2	Embedding density results	33
6.3	Stress-path results	35
6.4	Lifted-path results	38
6.5	Robustness of parameters	39

7	Markov bridges	44
7.1	Markov bridge motivation and background	44
7.2	Markov bridge methodology	44
7.3	Markov bridges results	46
8	Hidden Regimes on the Macroeconomic Manifold	48
9	PCA benchmark results	50
10	Conclusions and outlook	51
	Appendices	53
A	First appendix	53
B	Second appendix	54

List of Tables

1	Minimum degree and degree CV across bandwidths.	32
2	Percent of the near-collision pairs that are no longer in the closest $x\%$ of pairs in the higher-dimensional embedding. Near-collisions are pairs are the top 1% of closest nodes in the lower-dimensional embedding after applying a twelve-month temporal exclusion window.	33
3	Network Metrics by k	34
4	Macroeconomic Regimes and Event Analysis	35
5	Stress-path evaluation across the four historical scenarios. Higher values of $M_{\min}$ and M_{mean} indicate greater historical support. L_{diff} is measured in diffusion-coordinate space and D_{turn} is the total turning angle.	37
6	Model Performance Metrics	38
7	Nearest-neighbour diagnostics for the lifted linear paths. Distances are measured in the quantile-standardised macroeconomic space. The ratio R_{95} compares the 95th percentile of the path distances with the 95th percentile of the leave-one-out nearest-neighbour distances among the historical observations. Values of $R_{95} > 1$ indicate that the lifted path contains points farther from the historical data cloud than is typical at the corresponding historical threshold.	38

List of Figures

1	Spectrum of the transition matrix for different bandwidths.	32
2	Diffusion-map representation of the quantile-transformed macroeconomic observations using $\alpha = 1, \varepsilon = 3$. Plotted against different coordinates and coloured by different variables to show the different structures found by different coordinates.	34
3	Plot of the density penalty of diffusion coordinates (V) over time with NBER recession dates shaded grey. We also highlight points with high density penalties, but not in NBER recession dates.	35
4	Caption	36
5	Caption	37
6	Lifted macroeconomic trajectories for the early-1980s recession and the Global Financial Crisis. The graph path consists of observed transformed macroeconomic states, whereas intermediate points on the linear diffusion-coordinate path are reconstructed using local neighbourhood lifting.	40
7	Lifted macroeconomic trajectories for the COVID-19 disruption and the 2022 monetary-tightening episode. The graph path consists of observed transformed macroeconomic states, whereas intermediate points on the linear diffusion-coordinate path are reconstructed using local neighbourhood lifting.	41
8	Caption	41
9	Caption	42
10	Caption	43
11	Caption	47
12	Caption	47
13	Caption	48
14	Caption	49
15	Lifted macroeconomic trajectories for the COVID-19 disruption and the 2022 monetary-tightening episode. The graph path consists of observed transformed macroeconomic states, whereas intermediate points on the PCA path are reconstructed pseudo-inverse of PCA map.	54

1 Introduction

James thinks the intro might have too many sections. Maybe not all necessary.

1.1 motivation

Financial stress testing defines a methodology which aims to understand “the extent to which a portfolio, individual institution, or a group of institutions is capable of withstanding external shocks” [2]. For example, a bank may ask a fund manager “what will happen to your portfolio if we experience an economic shock similar to COVID-19 next year”. The purpose of this is to ensure that individuals and institutions are able to stay afloat during times of economic turmoil; as opposed to defaulting and needing bailed out. While the specific economic stress point is important, what is potentially equally important is the path the economy takes to get there. This is because in many portfolios and institutions, the final value may change along the way, due to risk limits or margin calls which would potentially mean a portfolio would have to be rebalanced.

What is the current method of generating stress paths? Is this even a thing?

The question then becomes, how do we generate economically plausible stress paths? Before we tackle this question, we introduce something called the manifold hypothesis. We assume that all of the observed variables in the economy are driven by a smaller number of underlying, unseen factors. This then suggests that the observed economic variables may concentrate near a lower dimensional structure. This motivates the use of dimensionality reduction schemes to try to describe the economy in a smaller number of dimensions. Linear dimension reduction methods, like Principle Component Analysis (PCA), assume that the data can be described by a linear subspace. PCA identifies the linear direction of maximal variance and uses the leading maximal directions of variance to generate a lower-dimensional subspace which can capture a lot of the variance in the data. A problem with this is that the relationship between macroeconomic variables, and themselves and the underlying factors may not be linear. This motivates the use of diffusion maps for a dimension reduction scheme, which was introduced by [4] as a “geometrically faithful dimensionality reduction technique” [1].

Even when using diffusion maps, the challenge of generating stress paths is not trivial. In [1], they use predefined stress paths for a subset of the macroeconomic variables to find a conditional distribution on the rest of the variables, and hence can generate stress paths. This is done by running a Kalman filter on the diffusion coordinates. In this dissertation, we intend to find a way to interpolate between two endpoints without any information about the interior points.

- What is stress testing?
- Introduce the manifold hypothesis
- Method reduction methods
- Diffusion maps
- In [1] they use linear paths in diffusion space, we want to beat that
- What can this be used for?

Limitations of linear interpolation through Euclidean space; it gives unrealistic combinations of variables.

Incorporate JDKF into motivation - argument runs that Baker has diffusion maps but linearly interpolates; can we find density aware paths

Each month gives many observations, but these could be represented using far fewer dimensions.

- learn low-dimensional representation of macro-financial states;

- construct stress transitions that respect the learned geometry and historical density;
- determine whether temporal regimes emerge in this latent representation.

1.2 Research questions

In this dissertation, we first try to find out whether diffusion maps can be used to recover stable and interpretable low-dimensional representations of macroeconomic-financial states. Then using this coordinate system, can we develop a novel method for interpolating between two endpoints in diffusion space whose points contain macro-financial combinations which are highly historically supported? The interpretation of these trajectories would be a stress path which passes through regions of macro-financial space which are highly supported by historical observations.

Main questions:

- Can diffusion maps recover a stable and interpretable low-dimensional representation of macro-financial conditions?
- Do graph-based and density-aware stress paths remain in more historically plausible regions than direct linear interpolation?

Extension questions:

- Do hidden Markov regimes fitted in diffusion space correspond to recognised economic regimes, including NBER recessions?
- Do density-aware geodesics pass through the inferred regimes in a temporally and economically coherent way?

1.3 contributions

In this dissertation, we construct a diffusion map from monthly macroeconomic and industry performance observations going back to 1964. We interpret each diffusion coordinates economic interpretation using non-linear distance correlation metrics and economic reasoning. We estimate historical support for diffusion coordinates via a density proxy. Using the density proxy, we introduce a density-aware graph cost for constructing stress transitions. We compare our density-aware path with linear interpolations **I think this will need to be explain/justified more above. Linear interpolation makes sense in PCA space because everything is linear and the space is defined. Whereas diffusion “space” is a concept, we have got a discrete approximation of it and unpenalised graph paths across four historical scenarios. We additionally employ a define a lifting procedure which allows for diffusion coordinate paths to be interpreted as macroeconomic variables.**

Have done:

- Implementation and numerical validation of a diffusion-map pipeline;
- Construction of density-aware graph geodesics for macroeconomic stress testing;
- Lifting of latent stress paths into economically interpretable variables

Aim to do:

- Fitting of a Gaussian HMM to diffusion coordinates rather than the original high-dimensional observations;
- Joint evaluation of geometric paths, historical density, and temporal regimes.

1.4 Findings

We find that the density-aware paths generally improve the minimum and weighted mean historical support, compared to linear interpolation in diffusion space and ordinary graph paths.

2 Background

2.1 High-dimensional state vectors and the manifold hypothesis

Here we introduce the notation used throughout this dissertation and expand on the manifold hypothesis discussed in Section 1. We define a macroeconomic-financial state variable as

$$z_t = [z_t^1, \dots, z_t^D]^\top \in \mathbb{R}^D,$$

which represents a variety of different macroeconomic and industry variables at time t . For example, z_t^1 may represent the unemployment rate and z_t^D may represent the previous month returns of a financial asset at time t . Although D may be very large, the collection of observed states may concentrate near a lower-dimensional set

$$\mathcal{M} \subset \mathbb{R}^D,$$

where $\dim(\mathcal{M}) \ll D$. In this dissertation, we assume the existence of some sort of non-linear manifold $\mathcal{M}$, but it is never explicitly calculated or estimated. The intrinsic geometry of $\mathcal{M}$ is learned via the diffusion mapping. citation for this theory. where did it come from? Can I just explain it well enough as a theory? The Manifold Hypothesis is Assumption 2 of [1]. In their introduction, they just say that they assume that the covariates and responses $(x(t), y(t))$ are nonlinear functions of a true, possibly lower-dimensional, underlying state of the system $\theta_t \in \mathbb{R}^d$. We don't ever use these θ_t explicitly, so is it worth introducing them rather than just explaining the concept?

Do I need a link to the next part of the background to this part? Like I did in my intro? I think right now they seem slightly disconnected. Should I say "The fact that the manifold may be non-linear motivates us to use diffusion maps which we outline below"?

2.2 Latent diffusions and spectral coordinates

I don't think that this part is necessary. Maybe this is where you can explain how the eigenvalues approach the eigenfunctions of the Laplace-Beltrami operator on the underlying manifold. Explanation of the latent process and how the diffusion map is suitable for this model. This is heavily based on the JDKF paper.

Assumed latent process:

$$d\theta_t = -\nabla U(\theta_t)dt + \sqrt{2}dW_t,$$

and its infinitesimal generator:

$$\mathcal{L} = \Delta - \nabla U \cdot \nabla.$$

Generator eigenfunctions provide dynamically meaningful coordinates. The essential relation is

$$\psi_j(z_t) \approx \varphi_j(\theta_t)$$

where ψ_j is obtained from the finite-data graph and φ_j is an eigenfunction of the limiting differential operator.

2.3 Diffusion maps

This outlines how we actually build the diffusion maps, rather than the theory above. Include an example of what diffusion coordinates actually do

Diffusion maps are a nonlinear dimensionality-reduction technique which use local similarities between observations to recover the geometry of a data set [4]. Suppose that

$$\mathcal{Z} = \{z_1, \dots, z_N\} \subset \mathbb{R}^D$$

is a collection of observations sampled from, or concentrated near, a lower-dimensional manifold

$$\mathcal{M} \subset \mathbb{R}^D.$$

The method constructs a weighted graph on the observations and uses the spectral properties of a random walk on this graph to obtain intrinsic coordinates for the data.

The first step is to calculate the pairwise squared Euclidean distances

$$\delta_{ij} = \|z_i - z_j\|^2.$$

These distances are converted into affinities using the Gaussian kernel

$$W_{ij} = \exp\left(-\frac{\|z_i - z_j\|^2}{\varepsilon}\right), \quad (2.1)$$

where the bandwidth $\varepsilon > 0$ controls the scale over which two observations are regarded as neighbours. **Mention the fact that this is different to Mahalanobis distance used in [1]. Mahalanobis is not sensitive to nonlinearity and correlation. You have to estimate the correlations to use this; with $\hat{Z} \in \mathbb{R}^{N \times D}$ this would introduce errors. Using the quantile data, we don't have to worry about the nonlinearity because it makes everything Gaussian, but we need to explain the correlation thing.** The matrix

$$W = (W_{ij})_{i,j=1}^N$$

is symmetric and contains values between zero and one. Nearby observations have affinity close to one, whereas the affinity between distant observations is close to zero. Since each observation has zero distance from itself, $W_{ii} = 1$.

The weighted degree of observation i is defined by

$$q_i = \sum_{j=1}^N W_{ij}. \quad (2.2)$$

Unlike the degree of an unweighted graph, q_i does not simply count the number of adjacent vertices. It measures the total affinity of z_i to the sample. Consequently, a large value of q_i indicates that z_i has many nearby observations, or several observations to which it has strong affinity. Conversely, a value close to one indicates that almost all of the degree is contributed by the diagonal entry $W_{ii} = 1$, and hence that z_i is nearly isolated at the chosen bandwidth. The degree therefore acts as a kernel-based measure of local sampling density.

If the observations are sampled non-uniformly from $\mathcal{M}$, densely sampled regions may dominate the resulting random walk. Diffusion maps allow this effect to be controlled through the density-normalised affinity matrix

$$W_{ij}^{(\alpha)} = \frac{W_{ij}}{q_i^\alpha q_j^\alpha}, \quad \alpha \in [0, 1]. \quad (2.3)$$

The parameter α determines how strongly the empirical sampling density is removed. The choice $\alpha = 0$ leaves the original affinities unchanged, whereas $\alpha = 1$ removes the leading influence of non-uniform sampling density and, under standard asymptotic conditions, yields an approximation to the Laplace–Beltrami operator associated with the underlying manifold [4]. Intermediate choices retain some dependence on the sampling distribution.

A second degree vector is then formed from the normalised affinities:

$$d_i^{(\alpha)} = \sum_{j=1}^N W_{ij}^{(\alpha)}. \quad (2.4)$$

Writing

$$D_\alpha = \text{diag} \left(d_1^{(\alpha)}, \dots, d_N^{(\alpha)} \right),$$

the diffusion transition matrix is

$$P = D_\alpha^{-1} W^{(\alpha)}, \quad P_{ij} = \frac{W_{ij}^{(\alpha)}}{\sum_{k=1}^N W_{ik}^{(\alpha)}}. \quad (2.5)$$

Each row of P sums to one, so P can be interpreted as the transition matrix of a Markov chain on the data. In particular, P_{ij} is the probability of moving from observation z_i to observation z_j in one step. The entries of P^t give the corresponding transition probabilities after t steps.

Although P is generally not symmetric, it is similar to the symmetric matrix

$$S = D_\alpha^{-1/2} W^{(\alpha)} D_\alpha^{-1/2}. \quad (2.6)$$

I think I should leave this section out of the background. This is just an implementation choice which goes in methodology Indeed,

$$P = D_\alpha^{-1/2} S D_\alpha^{1/2}.$$

The matrices P and S therefore have the same eigenvalues. Since S is symmetric, its eigenvectors are orthogonal and its eigendecomposition is numerically more stable than directly diagonalising P . If

$$S u_k = \lambda_k u_k,$$

then the corresponding right eigenvector of P is

$$\phi_k = D_\alpha^{-1/2} u_k.$$

The eigenvalues are ordered as

$$1 = \lambda_0 \geq \lambda_1 \geq \lambda_2 \geq \dots,$$

where the first eigenvector ϕ_0 is constant and carries no geometric information. The diffusion-map embedding at diffusion time t is therefore defined using the leading non-trivial eigenpairs:

$$\Psi_t(z_i) = (\lambda_1^t \phi_1(i), \dots, \lambda_r^t \phi_r(i)) \in \mathbb{R}^r. \quad (2.7)$$

The eigenvalue factors suppress rapidly decaying modes, while the eigenvectors describe the dominant directions of diffusion through the data. With the complete eigendecomposition, Euclidean distance in diffusion-coordinate space is equal to diffusion distance:

$$D_t^2(z_i, z_j) = \sum_{k \geq 1} \lambda_k^{2t} (\phi_k(i) - \phi_k(j))^2.$$

Truncating this expansion after r terms provides a low-dimensional approximation that retains the most persistent diffusion modes. How does this relate to the geodesic distances?

Under suitable assumptions on the sampling process, together with an appropriate relationship between the sample size and the bandwidth, the graph-based operator converges to a differential operator on $\mathcal{M}$. Correspondingly, its eigenvectors converge to evaluations of the eigenfunctions of the limiting operator at the sampled observations. This provides the theoretical basis for interpreting diffusion coordinates as empirical approximations to intrinsic spectral coordinates of the underlying system. In the latent-dynamical setting considered in Section 2.2, this connection motivates the use of diffusion maps to obtain data-driven coordinates for an otherwise unobserved state space.

The bandwidth ε is central to the construction. If ε is too small, the off-diagonal affinities

become negligible, so that

$$W \approx I$$

and many observations are effectively isolated. In this regime, the diffusion operator describes a poorly connected collection of local components. If ε is too large, most affinities approach one, so that

$$W \approx \mathbf{1}\mathbf{1}^\top,$$

and the kernel loses its ability to distinguish local geometric structure. An appropriate bandwidth must therefore be sufficiently large to connect the observations while remaining sufficiently local to preserve the geometry of the data.

The following comes from [7] (Figure 3, Equation 26). Might be worth mentioning “A popular choice for practitioners is to choose ε as the median of all distances between data points” [1]. One diagnostic for the bandwidth is the kernel mass

$$T(\varepsilon) = \sum_{i,j=1}^N \exp\left(-\frac{\|z_i - z_j\|^2}{\varepsilon}\right). \quad (2.8)$$

For observations sampled from a sufficiently regular d -dimensional manifold, the kernel mass is expected to satisfy

$$T(\varepsilon) \approx C\varepsilon^{d/2}$$

over an appropriate range of local scales. It follows that

$$\frac{d \log T(\varepsilon)}{d \log \varepsilon} \approx \frac{d}{2}. \quad (2.9)$$

A stable region in the log–log slope can therefore provide information about both the intrinsic dimension and a suitable range of bandwidths. In finite, noisy, and non-uniformly sampled data, however, such a scaling region may be weak or absent. Kernel-mass scaling is therefore best treated as a diagnostic and considered alongside graph connectivity, the degree distribution, spectral stability and the quality of the resulting coordinates.

The following comes from [3], Subsection 3.2 The embedding dimension r determines how many non-trivial diffusion modes are retained. A common diagnostic is the presence of a spectral gap. If

$$\lambda_r - \lambda_{r+1}$$

is large relative to neighbouring gaps, the first r non-trivial eigenvectors represent comparatively persistent modes, whereas subsequent eigenvectors decay more rapidly and may primarily reflect fine-scale variation or noise. Spectral gaps do not provide an infallible choice of dimension, particularly when eigenvalues occur in near-degenerate groups. The final dimension should therefore also be assessed through stability under nearby bandwidths, preservation of relevant geometric structure, and the requirements of the downstream analysis.

Include the example here? I am not sure how the example will exactly fit in.

2.4 Kernel scale and spectral diagnostics

This is included above Explanation of the theory behind choosing our model parameters basically. The kernel mass plots, spectral analysis, and the justification of the embedding dimension. Also graph connectivity and robustness of all choices.

- kernel mass $T(\varepsilon)$;
- the scaling relation $T(\varepsilon) \propto \varepsilon^{d/2}$;
- spectral gaps;
- eigenvalue multiplicities;

- graph connectivity;
- stability across nearby parameter values.

2.5 Graph geometry and shortest paths

The diffusion coordinates are a way to represent our data in a lower-dimensional structure. And as shown above, the short scale distances in diffusion space highly correlate to distances in the intrinsic geometry. I.e. for short distances, diffusion distance and geodesic distances are tied together. This leads us to wonder whether paths through diffusion coordinates via short jumps correspond to geodesic paths between points in the underlying geometry.

We therefore construct k NN graphs from the diffusion coordinates as an approximate representation of our manifold **I am not sure if this is complete. It is an approximation of the diffusion coordinate manifold, I think, but not the real manifold.** Once we have a fully connected graph **fully connected != complete right?**, we can use a shortest path algorithm to find a path between any two nodes in our diffusion coordinates, which then will be made up of short paths corresponding to geodesics on the intrinsic dimension. We therefore say that the shortest graph path represents a geodesic along the intrinsic data.

We demonstrate this in an example, continued from the previous section. **SHOW GRAPH PATHS IN EXAMPLE DIFFUSION COORDINATE AND THE DISTANCE CORRELATION WITH GEODESIC FOR ALL PAIRS, NOT JUST CLOSE PAIRS. OR LIFTED POINTS IN THE GRAPH PATH INTO THE AMBIENT SPACE, SHOWING THE PATH FOLLOWS THE MANIFOLD.**

Then we can see that graph paths through diffusion coordinates correspond to geodesics in the intrinsic geometry. **Not sure how possible it is to make this claim. The toy model results look good, but idk if this is real**

“The graph distance d_G is a good approximation for the geodesic distance d in the original data and in any isometric embedding, as it will closely follow the actual manifold rather than cross in the ambient space.” [6]

- Graph paths;
- ordinary graph-geodesic cost;
- the distinction between a Euclidean chord and a path constrained to observed neighbourhoods;
- Dijkstra’s shortest-path problem.

2.6 Density-aware graph geometry??

Maybe worth going into extra detail here because this is novel

In this dissertation, we define a proxy for historical support as the density of diffusion coordinates. **reference the section?** While the shortest graph paths represent the approximate shortest paths along the learned geometry of the data, we would like to traverse along the learned geometry such that the points we visit have high historical support.

To find the shortest paths, we use Dijkstra’s algorithm, which includes the edge cost function

$$w_{ij} = \|x_i - x_j\|,$$

where the algorithm finds a path between nodes by minimising the sum of the edge costs. We introduce a modified edge cost which incorporates a sparsity penalty.

Suppose that ρ_i is the density of a node i in diffusion coordinates **I want to say diffusion space, but I haven’t defined that and I don’t know if I will**, then we can define a density potential

as $V_i = -\log(\rho_i)$, which is high for low-density nodes and low for high-density nodes. Then we define the density-aware edge cost as

$$w_{ij}^{(\beta)} = \|x_i - x_j\| \exp\left(\beta \frac{V_i + V_j}{2}\right),$$

where β controls the strength of the sparsity avoidance. We can see that setting $\beta = 0$, we simply have the standard edge cost. However, increasing β means that the cost of an edge that involves a low-density node (high V_i) is now higher than it was previously. Depending on the density measure, it may be possible for points to have a density $\rho_i > 1$, which would correspond to a $V_i < 0$. This would then mean that an edge containing two high-density nodes (at least, both nodes have $\rho_i > 1$), would have a smaller edge cost with the addition of the exponential term. This has the effect of making higher-density nodes more attractive to the algorithm.

- regions visited frequently by history correspond to high density
- low density means combinations that have not been seen before.
- show the negative correlation with distance and density.

2.7 Hidden Markov models??

This is the extension, and I am not sure if it will be done

3 Data and Preprocessing

3.1 Data sources and sample period

In this dissertation, we use two data sources to create our training data matrix. We use macroeconomic variable data from the Federal Reserve Economic Database (FRED) and portfolio returns data from the Kenneth R. & French Library. Additionally, we use data on economic recessions provided by the National Bureau of Economic Research (NBER) for downstream analysis.

The macroeconomic data for this dissertation was the largest dataset we used and it also required the most amount of work done to it. We downloaded the 2026-05 FRED-MD dataset, which was the most recent version at the time. This dataset contained monthly data going back to January 1959 for 126 macroeconomic variables for the United States economy. After removal of redundant variables, removal of missing values and applying transformations, we end up using the data from August 1963 for 35 macroeconomic variables. Along with each variable, FRED provide transform codes to make the variables stationary.

Motivated by [1], we also include data on the returns of assets in the U.S. over the same time period. **Is this a good enough motivation?** We include 10 separate portfolios each comprising of assets from different industries. The Kenneth R. and French Library provides returns data on the 10 separate portfolios going back to July 1926 until May 2026. We therefore use the returns data from August 1963 to match the FRED-MD dataset. The portfolios are constructed every year, grouping each asset into a specific industry. This avoids the survivorship bias which would be found if we used individual asset returns. **Why? Citation? anymore needed?** The dataset contains the rates of returns for each portfolio, so therefore there are no transforms needed as the data is already assumed to be stationary **Is this fine to assume/say?**

Finally, for downstream analysis, we use a monthly recession indicator using dates from the NBER.

3.2 Variable selection

The FRED splits up the different variables into different groups which represent different areas of the economy. For example, Group 6 is named “Interest and exchange rates” which contains 22 different macroeconomic variables to do with the interest and exchanges rates in the U.S. Including all of these variables would introduce a lot of redundancies, for example there is a variable “AAAFFM” which gives the Moody’s tripple A rated corporation yield rate minus the Federal Reserve’s interest rate, and there is also both “AAA” and “FEDFUNDS”. Including all three of these variables would not add any new information when compared to adding any two of them. We also have variables which are very highly correlated which may introduce more noise than new information. The diffusion map uses Euclidian distances to create an affinity matrix. Suppose that we have two very similar standardised variables

$$x_{1,t} \approx x_{2,t}.$$

Then their contribution to the squared distance is then approximately

$$(x_{1,i} - x_{1,j})^2 + (x_{2,i} - x_{2,j})^2 \approx 2(x_{1,i} - x_{2,j})^2,$$

so the addition of the correlated variable may mostly distort specific directions in the geometry instead of adding new information. We then choose a representative sample from each of the 8 groups with the aim of capturing as much of the economic description as possible. **explain how you did this, maybe? Looked at correlations, looked at definitions of variables, etc.** This left us with 28 variables from the FRED-MD dataset, combined with the 10 industry portfolios; we have 38 different variables.

3.3 Temporal alignment and missing observations

All of the data sources provided the monthly data from the first date of every month, so there was no need to align the data in any special way. In the entire combined dataset which contained over 28,000 data points, there were 4 missing values. This is due to a government shutdown in 2025 where some variables were not calculated by FRED **reference, or do I even need to include this? Seems like a nice detail to know and slightly interesting, but not related to the diss at all.** Since the missing values only happened once for only 4 variables over 753 observations at the same date, we deemed it appropriate to use linear interpolation to fill in the missing values.

3.4 Transformations

We apply transformations to the FRED-MD dataset to make the series stationary **why do you do this?**. FRED provide recommended transformations to apply to their variables which we mostly use. Initially, the recommended transformatoins were used, however, this caused problems downstream where the diffusion coordinates would only act as indicator variables for the COIVID-19 shock. This is because for many variables, the COVID-19 shock was orders of magnitude greater than any shock seen before. This was especially significant for activity variables such as total employment (“PAYEMS”) when we used month-on-month transformations. To remedy this, we used year-on-year transformations for a selection of variables. This is because these activity variables were know to be cyclic, and so yearly transformations still made economic sense and it dampened the impact of very large shocks like COVID-19. Using year-on-year transformations means that we have 12 fewer observations post-transformation.

Additionally, due to the COVID-19 shock, the unemployment rate and the number of unemployment claims were also massive spikes. We changed these to just be levels instead of difference. This resulted in both the series still being stationary, while having a lower kertosis - meaning the spike was less of an outlier; closer to the rest of the data points.

Maybe a plot of the difference between MoM and YoY or first difference and levels

3.5 Standardisation

While the change of transformations described above improved the downstream diffusion coordinate results, the same problem still persisted without standardisation. We used standardisation to make all of the variables of similar scales. Many standardisation methods were experimented with, but we ended up using quantile standardisation. This standardisation technique prevents heavy tails by forcing each variable to follow a normal distribution, while keeping the ordering of the observations within each variable preserved. **this might not be enough explanation.** This fixes the problem with the diffusion coordinates only showing a massive spike for COVID. It makes the geometry less sensitive to outliers. It does this by making everything follow the same distribution. A drawback is, however, that the distance is not preserved within individual variables. This means that the model may struggle to differentiate between a severe crisis and a mild crisis. This worry is negated slightly because we not not so interested in the single values of the variables, but the joint values, and in a severe crisis state, many values are expected to be in the extreme quantiles, so this can still result in diffusion coordinates which are far apart. Quantile standardisation can also distort the linear relationships between variables, so for example Pearson correlations may change before and after the standardisation. So we use the distance correlation for our correlation analysis which isn’t as impacted by this **Citation / is it even true?**. Additionally, forcing every variable to have approximately the same distribution may mean that noisy and economically less significant variables to have the same marginal influence as highly informative series **Would this not be a problem for all standardisation methods?**. This is why it is important to correctly identify the important variables as discussed above.

Finally, the standardisation being dependent on all samples means that if we add a new sample to the data, it may change the whole standardisation. This may lead to look-ahead bias in out-of-sample testing and would therefore need to be more carefully considered in other works.

3.6 Final analysis matrix

The final working data matrix is

$$\tilde{Z} \in \mathbb{R}^{743 \times 38},$$

consisting of 743 monthly observations of 38 economic variables.

4 Numerical methods

4.1 Overview of the computational pipeline

This is where you give a clear picture of what you actually are going to do. Reference the different sections, etc.

raw data $\rightarrow$ transformation $\rightarrow$ diffusion map $\rightarrow$ density and graph $\rightarrow$ stress paths $\rightarrow$ lifting

? $\rightarrow$ HMM regime analysis

In this section, we describe the entire computational pipeline. Firstly, using our transformed and standardised data, we find a diffusion map, which captures the potential non-linear structure of the macroeconomic manifold on which the data lies. Using that diffusion map, we choose an appropriate embedding dimension, and now we have our new coordinate system. Using these new coordinates, we construct a k -nearest-neighbours graph and compute an estimation of the densities. We then implement our density-aware graph path model to interpolate a path between a variety of specifically chosen economic endpoints. We compare this path to various other types of paths, specifically, a linear path in diffusion space, and a standard graph path. Then we can lift these paths back to the macro space, and there we can compare different paths in real coordinates to see how well our model does.

I'm not sure if I should include the above section yet, so I haven't worked on it. You can skip this part and start with Section 4.2.

4.2 Diffusion-map construction

The diffusion-map construction introduced in Section 2.3 was applied to the transformed macroeconomic state matrix

$$\tilde{Z} = \begin{pmatrix} \tilde{z}_1^\top \\ \vdots \\ \tilde{z}_N^\top \end{pmatrix} \in \mathbb{R}^{N \times D},$$

where each row $\tilde{z}_i \in \mathbb{R}^D$ represents one monthly macroeconomic observation and each column corresponds to one of the variables described in Section 3. The NBER recession indicator was retained for subsequent external comparison, but was not included as an input to the embedding.

The construction follows the diffusion-map framework from Section 2.3, but is restated here in computational form to make explicit how the empirical embedding was obtained. Pairwise squared Euclidean distances were first calculated as

$$\delta_{ij} = \|\tilde{z}_i - \tilde{z}_j\|^2, \quad i, j = 1, \dots, N.$$

These distances were converted into Gaussian affinities,

$$W_{ij} = \exp\left(-\frac{\delta_{ij}}{\varepsilon}\right),$$

where $\varepsilon > 0$ controls the neighbourhood scale represented by the kernel.

The kernel degrees

$$q_i = \sum_{j=1}^N W_{ij}$$

were then used to construct the density-normalised affinity matrix

$$W_{ij}^{(\alpha)} = \frac{W_{ij}}{q_i^\alpha q_j^\alpha}.$$

James asked why we don't take $\sqrt{\cdot}$ here Throughout the main empirical analysis, the density-

normalisation parameter was fixed at

$$\alpha = 1.$$

As discussed in Section 2.3, this choice removes the leading influence of non-uniform sampling density from the diffusion operator and is therefore appropriate when the objective is to recover the geometry represented by the observations rather than to preserve their empirical sampling frequency.

The row sums of the normalised affinity matrix are

$$d_i^{(\alpha)} = \sum_{j=1}^N W_{ij}^{(\alpha)},$$

and define the diagonal matrix

$$D_\alpha = \text{diag} \left(d_1^{(\alpha)}, \dots, d_N^{(\alpha)} \right).$$

The corresponding diffusion transition matrix is

$$P = D_\alpha^{-1} W^{(\alpha)}.$$

why? This is just by definition, I think; doing this means that the row sums are equal to 1, so then each element corresponds to the probability as desired. This is the Markov operator whose right eigenvectors define the diffusion coordinates introduced in Section 2.3.

The highlighted red pattern is seen a lot and James says it's an LLM tell. Don't use this In the numerical implementation, however, the eigenpairs were not obtained by directly diagonalising the generally non-symmetric matrix P . Instead, the symmetric matrix

$$S = D_\alpha^{-1/2} W^{(\alpha)} D_\alpha^{-1/2}$$

was constructed. Since

$$P = D_\alpha^{-1/2} S D_\alpha^{1/2},$$

the matrices P and S are similar and therefore have the same eigenvalues. The symmetry of S allows its eigenpairs

$$S u_k = \lambda_k u_k$$

to be computed using a symmetric eigensolver. The corresponding right eigenvectors of the diffusion transition matrix are then recovered through

$$\phi_k = D_\alpha^{-1/2} u_k.$$

Indeed,

$$P \phi_k = D_\alpha^{-1/2} S u_k = \lambda_k \phi_k,$$

so the recovered ϕ_k are precisely the eigenvectors entering the diffusion-map representation.

The eigenpairs were ordered by decreasing eigenvalue,

$$1 = \lambda_0 \geq \lambda_1 \geq \lambda_2 \geq \dots,$$

and the trivial eigenpair (λ_0, ϕ_0) What does this pair represent? This is the smoothest way to assign coordinates, but it gives no information. Can maybe talk about because we normalised P , then we have that the biggest eigenvalue corresponds to the constant eigenvector... or choose a response which talks about whether this represents an intercept? Idk if it does though was removed. At diffusion time $t = 1$, retaining the first r non-trivial modes gives

$$\Psi(\tilde{z}_i) = (\lambda_1 \phi_1(i), \dots, \lambda_r \phi_r(i)) \in \mathbb{R}^r.$$

James was asking whether this was PCA in D_α inner product - not really sure. I think that diffusion maps are a special case of something called “kernel PCA” (not sure if that’s the name) Equivalently, collecting the coordinates for all observations gives the embedding matrix

$$\Psi = [\lambda_1 \phi_1 \quad \cdots \quad \lambda_r \phi_r] \in \mathbb{R}^{N \times r}.$$

The complete computational procedure is summarised in Algorithm 1.

Algorithm 1 Diffusion-map embedding

- 1: **Input:** $\tilde{Z} \in \mathbb{R}^{N \times D}$, ε , α , r
 - 2: **Output:** Embedding coordinates $\Psi \in \mathbb{R}^{N \times r}$
 - 3: $\delta_{ij} \leftarrow \|\tilde{z}_i - \tilde{z}_j\|^2$
 - 4: $W_{ij} \leftarrow \exp(-\delta_{ij}/\varepsilon)$
 - 5: $q_i \leftarrow \sum_{j=1}^N W_{ij}$
 - 6: $W_{ij}^{(\alpha)} \leftarrow W_{ij}/(q_i^\alpha q_j^\alpha)$
 - 7: $d_i^{(\alpha)} \leftarrow \sum_{j=1}^N W_{ij}^{(\alpha)}$
 - 8: $D_\alpha \leftarrow \text{diag}(d_1^{(\alpha)}, \dots, d_N^{(\alpha)})$
 - 9: $P \leftarrow D_\alpha^{-1} W^{(\alpha)}$
 - 10: $S \leftarrow D_\alpha^{-1/2} W^{(\alpha)} D_\alpha^{-1/2}$
 - 11: Compute $Su_k = \lambda_k u_k$
 - 12: $\phi_k \leftarrow D_\alpha^{-1/2} u_k$
 - 13: Sort (λ_k, ϕ_k) in descending order of λ_k
 - 14: Discard the trivial eigenpair (λ_0, ϕ_0)
 - 15: $\Psi \leftarrow [\lambda_1 \phi_1, \dots, \lambda_r \phi_r]$
 - 16: **return** Ψ
-

Algorithm 1 requires two choices that materially determine the resulting representation: the kernel bandwidth ε , which controls the scale at which local observations are connected, and the retained dimension r , which determines the number of non-trivial diffusion modes used to represent the data. Rather than fixing these quantities a priori, both were assessed empirically using the diagnostics described in the following sections. The selected values were then held fixed for the principal graph construction, density estimation and stress-path analysis, with sensitivity to these choices examined separately in Section 6.5.

4.3 Diffusion parameter selection

The kernel bandwidth ε controls the locality of the Gaussian affinity kernel and was selected before fixing the embedding dimension. As an initial scale diagnostic, the kernel mass

$$T(\varepsilon) = \sum_{i,j} \exp\left(-\frac{\|\tilde{z}_i - \tilde{z}_j\|^2}{\varepsilon}\right)$$

was evaluated over a logarithmically spaced range of bandwidths, together with its local log–log slope, as described in Section 2.4. This diagnostic identifies scales at which the total kernel mass changes most rapidly **how?**. However, in a finite and non-uniformly sampled dataset, the bandwidth selected from the slope alone may produce a poorly connected graph **why?** **This might be something worth discussing; a mini result. I may also discuss all of this in the background bit.** The kernel-mass rule was therefore used to identify a plausible bandwidth range rather than as an automatic selection criterion.

Candidate bandwidths were subsequently compared using the connectivity and spectral prop-

erties of the resulting diffusion operator. Let

$$q_i(\varepsilon) = \sum_j W_{ij}(\varepsilon)$$

denote the unnormalised kernel degree of observation i . The minimum degree,

$$q_{\min}(\varepsilon) = \min_i q_i(\varepsilon),$$

was used to detect observations that were effectively isolated from the remainder of the sample. Since $W_{ii} = 1$, a value of q_i close to one indicates that almost all of the degree is contributed by the self-affinity. The heterogeneity of the degree distribution was measured using

$$\text{CV}_q(\varepsilon) = \frac{\text{sd}\{q_i(\varepsilon)\}}{\text{mean}\{q_i(\varepsilon)\}}.$$

A large coefficient of variation indicates that the graph connectivity is concentrated unevenly across observations, while comparison across neighbouring bandwidths provides a check on the stability of the connectivity structure.

The leading non-trivial eigenvalues of the transition matrix P were also compared across candidate bandwidths. Particular attention was given to the stability of the dominant eigenvalues and the gaps between consecutive eigenvalues under small changes in ε . The corresponding diffusion coordinates were inspected to exclude bandwidths for which the leading eigenvectors were nearly constant over most observations but contained isolated sharp spikes, as this behaviour is indicative of weakly connected or nearly isolated graph components.

Combining these diagnostics, the bandwidth was fixed at

$$\varepsilon = 3.$$

This choice provided adequate graph connectivity while retaining sufficient locality in the kernel. The kernel-mass diagnostic, degree statistics and spectral comparisons supporting this choice are reported in Section 6.1.

The following paragraph was unclear. Why not use the full spectrum? Explain the compromises. We want to reduce the dimensions to make things simpler and reduce noise maybe. The embedding dimension was not selected from the eigenspectrum alone. Candidate dimensions were assessed using three complementary criteria: the leading non-trivial eigenspectrum, the separation of near-collision pairs and the reconstruction performance of the local-neighbour lifting procedure, outlined in Section 4.7.

The eigenspectrum was used to identify the number of dominant diffusion modes and to locate large gaps between consecutive non-trivial eigenvalues. Since a spectral gap does not by itself guarantee that the retained coordinates distinguish all economically different observations, an additional near-collision diagnostic was introduced. For an embedding of dimension r , pairs of observations separated by no fewer than twelve months were ranked according to their Euclidean distance in diffusion space. The closest 1% of eligible pairs were defined as near-collisions. Their distance ranks were then recomputed after adding coordinate $r + 1$. The proportion of these pairs that left the closest 1%, 5% or 10% in the higher-dimensional embedding measured how frequently the additional coordinate separated observations that had appeared nearly identical in the lower-dimensional representation. A large exit proportion therefore indicates that coordinate $r + 1$ contains substantial information not represented by the first r coordinates.

The ability of each candidate embedding to reconstruct the transformed macroeconomic observations was assessed through the local-neighbourhood lift. If $\widehat{Z}^{(r)}$ denotes the observations reconstructed from an r -dimensional embedding, reconstruction accuracy was summarised using

the root mean squared error and the proportion of variance recovered,

$$R_{\text{lift}}^2(r) = 1 - \frac{\|Z - \widehat{Z}^{(r)}\|_F^2}{\|Z - \overline{Z}\|_F^2},$$

why do you use this? What does it mean? where $\overline{Z}$ contains the variable-wise sample means. This diagnostic measures the extent to which the retained diffusion coordinates preserve information required to recover the original transformed variables.

Finally, low-dimensional projections were inspected after colouring observations by calendar time, recession status and selected macroeconomic variables. These visualisations were used as a qualitative check that the retained coordinates represented distinct and economically interpretable directions of variation, rather than as a standalone dimension-selection rule.

The combined diagnostics led to the selection

$$r = 4.$$

The fourth coordinate provided a material improvement in near-collision separation and lifted reconstruction relative to the three-dimensional embedding, whereas the incremental benefit of adding a fifth coordinate was comparatively small. The numerical comparisons and embedding visualisations are presented in Section 6.1.

4.4 Graph construction and density estimation

Following the construction of the diffusion embedding, an undirected k -nearest-neighbours graph was formed from the observed diffusion coordinates,

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \quad \mathcal{V} = \{1, \dots, N\}.$$

A directed edge from node i to node j was first introduced whenever ψ_j was among the k nearest neighbours of ψ_i in diffusion space **what is diffusion space? Make this clear**. The graph was then symmetrised by taking the union of the directed edge sets, so that $(i, j) \in \mathcal{E}$ whenever either observation selected the other as a neighbour. Consequently, the average degree of the resulting graph may be slightly greater than k . The Euclidean length assigned to each edge was

$$\ell_{ij} = \|\psi_i - \psi_j\|.$$

The nearest-neighbour search was implemented using Scikit-Learn’s `NearestNeighbors` class.

Candidate values of k were assessed using several complementary graph diagnostics. First, the graph was required to be connected, ensuring that a path existed between any selected pair of endpoints. All candidate values considered satisfied this condition. Second, the graph density

$$d_{\mathcal{G}} = \frac{2|\mathcal{E}|}{N(N-1)}$$

was monitored to ensure that the graph remained local and did not approach a complete graph as k increased.

The presence of unusually long edges within shortest paths was also examined. A systematic subsample containing every fifth node was formed, and shortest paths were calculated between every unordered pair of nodes in this subsample. For each path Γ , the statistic

$$R_{\text{edge}}(\Gamma) = \frac{\max_{(i,j) \in \Gamma} \ell_{ij}}{\text{median}_{(i,j) \in \Gamma} \ell_{ij}}$$

was recorded. A large value indicates that the path depends upon an edge whose length is substantially greater than its typical edge length. Such an edge may act as an artificial shortcut across a sparsely sampled region, causing the graph path to represent a non-local jump rather

than gradual movement through the sampled manifold **why is this bad? Is this not a part of the geometry? Why don't we want this? This should be made clear in the small example.** The distribution of this statistic was therefore compared across candidate values of k .

Finally, the resulting paths and their principal conclusions were required to remain stable under nearby choices of k **why do we need this to be stable for k ? Just explain this a bit more.** These diagnostics were used to select the baseline graph specification, while a broader sensitivity analysis with respect to k is reported in the robustness analysis.

A relative density score was then estimated from the observations in diffusion space. For an arbitrary point $\psi \in \mathbb{R}^r$, define

$$\hat{\rho}(\psi; h) = \sum_{j=1}^N \exp\left(-\frac{\|\psi - \psi_j\|^2}{h}\right),$$

where $h > 0$ is another bandwidth parameter which controls the locality of the density estimate. At an observed node, this gives

$$\hat{\rho}_i = \hat{\rho}(\psi_i; h).$$

Since only relative density is required for the path construction, the kernel score was normalised by its mean over the observed nodes,

$$\bar{\rho} = \frac{1}{N} \sum_{i=1}^N \hat{\rho}_i, \quad \rho(\psi) = \frac{\hat{\rho}(\psi; h)}{\bar{\rho}},$$

with $\rho_i = \rho(\psi_i)$.

Candidate values of h were compared according to whether the resulting density field distinguished well-supported and sparsely supported observations without becoming nearly constant or being dominated by isolated spikes **what are spikes? Setting h to a particular value results in only a very dense region being assigned high density, whereas the rest of the points are very sharply assigned low density. The gradient is sharp.** Stability under nearby bandwidth choices was also examined. The selected value and its sensitivity are reported with the empirical results and robustness checks.

For numerical stability, a small density floor $\delta > 0$ was imposed,

$$\rho_{i,\delta} = \max\{\rho_i, \delta\},$$

This impacts the normalisation discuss/change this and the density potential at node i was defined by

$$V_i = -\log(\rho_{i,\delta}).$$

Under this construction, observations surrounded by many nearby historical states have relatively low potential, whereas isolated observations have relatively high potential.

The density score is interpreted as a measure of *historical support under the geometry learned by the diffusion map* **how? This is kind of a result, so maybe just don't mention it here.** It does not by itself establish that a state is economically plausible or dynamically attainable. Its empirical relationship with recognised periods of economic stress and with proximity to historical observations in the transformed macroeconomic space is examined in the results.

I think I can be less conservative here; there is a link between real support and density. Depends on what I put in the background, but I think the above may be underselling it slightly

4.5 Stress-path construction

4.5.1 Endpoint selection

Four historically motivated start–stress endpoint pairs were considered. Both the start and stress endpoints were required to correspond to observed dates and hence to existing nodes of the diffusion graph **why? So we can test the method directly instead of having to deal with out**

of sample methods. Can compare actual paths too?. No out-of-sample extension was therefore required for the principal stress-path analysis.

The endpoints were selected to represent economically interpretable transitions from comparatively benign macroeconomic conditions to distinct forms of historical stress how? how were these selected? Looked at the individual macro variables and checked dates with news. The specific dates, scenario labels and summary economic characteristics are introduced in the empirical results. More detailed historical discussion is provided separately where required.

4.5.2 Path definitions

Three paths were constructed between each pair of endpoints: a linear path through diffusion space, an ordinary graph shortest path and a density-aware graph shortest path. Comparing these constructions separates the effect of restricting movement to the empirical graph from the additional effect of favouring historically well-supported regions.

Let i_{start} and i_{stress} denote the graph nodes corresponding to the selected endpoints. A graph path is represented by an ordered sequence of node indices

$$\Gamma = (i_1, \dots, i_L),$$

where

$$i_1 = i_{\text{start}}, \quad i_L = i_{\text{stress}}, \quad (i_m, i_{m+1}) \in \mathcal{E}.$$

For a specified set of positive edge costs w_{ij} , the shortest path solves

$$\Gamma^* = \arg \min_{\Gamma: i_{\text{start}} \rightarrow i_{\text{stress}}} \sum_{m=1}^{L-1} w_{i_m i_{m+1}}.$$

The optimisation was implemented using Dijkstra's Algorithm via SciPy's graph shortest-path routine.

The ordinary graph path assigns each edge its Euclidean diffusion-space length,

$$w_{ij}^{(0)} = \ell_{ij} = \|\psi_i - \psi_j\|.$$

This path is constrained to move through observed graph nodes but does not explicitly distinguish between densely and sparsely supported regions.

To incorporate historical support, the density-aware edge cost was defined for $\beta \geq 0$ by

$$w_{ij}^{(\beta)} = \|\psi_i - \psi_j\| \exp \left(\beta \frac{V_i + V_j}{2} \right).$$

The average of the endpoint potentials approximates the density potential encountered while traversing the edge. Since low-density nodes have larger values of V_i , edges passing through sparsely supported regions receive a higher cost. The corresponding density-aware path is

$$\Gamma_\beta = \arg \min_{\Gamma: i_{\text{start}} \rightarrow i_{\text{stress}}} \sum_{(i,j) \in \Gamma} w_{ij}^{(\beta)}.$$

When $\beta = 0$, this reduces to the ordinary graph shortest path. Increasing β places progressively greater emphasis on historical support relative to direct path length.

Candidate values of β were compared according to the trade-off between improved density support and the additional length or detour introduced into the path. Stability across nearby values was also examined. The retained specification is used for the principal analysis, while the sensitivity of the results to β is considered in the robustness analysis.

As an unconstrained benchmark, a straight line was constructed between the same endpoints in diffusion space,

$$\gamma_{\text{lin}}(s) = (1 - s)\psi_{\text{start}} + s\psi_{\text{stress}}, \quad s \in [0, 1].$$

this is the LLM pattern again highlighted in red The path was evaluated on an equally spaced grid of values of s . Unlike the graph paths, its intermediate points **need not** coincide with observed historical states and it may pass through regions containing little empirical support. **It therefore** provides a diffusion coordinate space linear benchmark against which the effects of graph restriction and density weighting can be assessed **This seems like a very weak baseline. You need to describe what the diffusion space is better for this ever to make sense.**

The graph and linear paths are compared using the path-length, density-support and macroeconomic evaluation measures defined in Section 4.6.

4.6 Stress-path evaluation

The stress paths are evaluated according to three complementary properties: the historical support encountered along the path, the distance travelled through diffusion space, and the regularity of the resulting trajectory. These quantities are considered jointly, since a path may avoid sparsely supported regions only by taking a substantially longer or more irregular route.

For evaluation, let a discretised path be written as

$$\gamma = (\gamma_0, \dots, \gamma_L),$$

where $\gamma_\ell \in \mathbb{R}^r$. For a graph path, the points γ_ℓ are observed diffusion coordinates ψ_{i_ℓ} , whereas for the linear benchmark they are interpolated points along the straight line defined in Section 4.5.

The principal evaluation criterion is the density encountered along the path. For an interpolated point γ_ℓ that does not coincide with an observed state, the density score is evaluated against the historical diffusion coordinates using the same kernel and bandwidth as in Section 4.4,

$$\hat{\rho}(\gamma_\ell) = \sum_{j=1}^N \exp\left(-\frac{\|\gamma_\ell - \psi_j\|^2}{h}\right).$$

This is normalised using the same full-sample scale

$$\bar{\rho} = \frac{1}{N} \sum_{i=1}^N \hat{\rho}(\psi_i),$$

giving

$$\rho(\gamma_\ell) = \frac{\hat{\rho}(\gamma_\ell)}{\bar{\rho}}.$$

A leave-one-out modification is used whenever the evaluated point corresponds to an observed graph node. For $\gamma_\ell = \psi_i$, its evaluation density is

$$\hat{\rho}_{-i}(\psi_i) = \sum_{\substack{j=1 \\ j \neq i}}^N \exp\left(-\frac{\|\psi_i - \psi_j\|^2}{h}\right), \quad \rho_{-i}(\psi_i) = \frac{\hat{\rho}_{-i}(\psi_i)}{\bar{\rho}}.$$

James warns that this may be unstable - what if there is a point within some $\delta \ll 1$? Maybe explain/fix, but you can also just check that no such points exist in the data. I am not currently sure how it is unstable. Removing the observation's self-contribution prevents graph nodes from receiving an automatic affinity of one purely because they belong to the historical sample. This leave-one-out adjustment is used only for path evaluation; the density field used to construct the density-aware graph remains that defined in Section 4.4. The same convention is applied to any observed endpoint appearing on the linear path. After evaluation, the density floor δ is applied as before.

Denote the resulting evaluation density at γ_ℓ by $\rho_{\text{eval}}(\gamma_\ell)$. The primary support metric is the minimum density,

$$M_{\min}(\gamma) = \min_{0 \leq \ell \leq L} \rho_{\text{eval}}(\gamma_\ell),$$

which identifies the least historically supported state encountered along the path. Since this metric is determined by a single point, the overall level of support is also summarised by the path-average density. To account for differences in path discretisation, this is weighted by the diffusion-space length of each segment,

$$M_{\text{mean}}(\gamma) = \frac{1}{L_{\text{diff}}(\gamma)} \sum_{\ell=0}^{L-1} \frac{\rho_{\text{eval}}(\gamma_{\ell}) + \rho_{\text{eval}}(\gamma_{\ell+1})}{2} \|\gamma_{\ell+1} - \gamma_{\ell}\|,$$

where

$$L_{\text{diff}}(\gamma) = \sum_{\ell=0}^{L-1} \|\gamma_{\ell+1} - \gamma_{\ell}\|$$

is the total path length in diffusion coordinate space. This provides a measure of the additional distance required for a path to remain in better-supported regions of the learned diffusion geometry.

Finally, path regularity is measured through changes in direction between successive segments. Define the unit direction vector

$$u_{\ell} = \frac{\gamma_{\ell+1} - \gamma_{\ell}}{\|\gamma_{\ell+1} - \gamma_{\ell}\|}, \quad \ell = 0, \dots, L-1.$$

The turning angle at an interior path point is

$$\theta_{\ell} = \arccos(u_{\ell-1}^{\top} u_{\ell}), \quad \ell = 1, \dots, L-1,$$

and the total direction change is

$$T_{\text{turn}}(\gamma) = \sum_{\ell=1}^{L-1} \theta_{\ell}.$$

A straight path has zero direction change, while larger values indicate a more tortuous trajectory through diffusion space. **This seems inconsistent, is the euclidian inner product meaningful? I just used it as a visual aid, because the 2d projections of the path may look straight even though they are very jagged in other dimensions. Again, you mention diffusion space and being ‘tortuous’ in it. Change this wording for one, and check that even makes sense.**

LLM pattern again The path constructions are therefore compared using minimum and average historical support together with path length in diffusion coordinate space and mean direction change. The objective **is not** to optimise any one of these quantities independently, **but to** assess whether the graph-based constructions achieve greater historical support without requiring excessive detours or irregularity. The empirical comparisons are reported in Section 6.3.

4.7 Lifting paths into macroeconomic space

The stress paths constructed in diffusion space must ultimately be expressed in terms of the transformed macroeconomic variables. For graph paths, no estimated inverse mapping is required. Each graph vertex corresponds to an observed diffusion coordinate ψ_i and therefore to its associated macroeconomic observation $\tilde{z}_i$. Hence, for a graph path

$$\Gamma = (i_1, \dots, i_L),$$

the corresponding macroeconomic path is obtained directly as

$$(\tilde{z}_{i_1}, \dots, \tilde{z}_{i_L}).$$

In contrast, the intermediate points of the linear benchmark generally do not correspond to observed states and must therefore be lifted from diffusion space back to macroeconomic space.

Two lifting procedures were considered. The first is a global linear inverse map, following

the approach of Baker et al. [1]. Let

$$\psi_i^{(c)} = \psi_i - \bar{\psi}, \quad \tilde{z}_i^{(c)} = \tilde{z}_i - \bar{z},$$

denote the centred diffusion and macroeconomic coordinates. The lifting matrix is obtained from

$$H = \arg \min_{B \in \mathbb{R}^{D \times r}} \sum_{i=1}^N \left\| \tilde{z}_i^{(c)} - B \psi_i^{(c)} \right\|^2,$$

Is this well posed? (it looks like it isn't) and an arbitrary diffusion-space point ψ is lifted according to

$$\hat{z}_{\text{lin}}(\psi) = H(\psi - \bar{\psi}) + \bar{z}.$$

Centring the variables before fitting and subsequently restoring $\bar{z}$ is equivalent to allowing the linear reconstruction to contain an intercept. This construction will be referred to as the Baker-style lift.

The second procedure is a local-neighbourhood lift (LNL). For a point ψ , let $\mathcal{N}_m(\psi)$ denote its m nearest historical observations in diffusion space. The lifted state is defined as the weighted average

$$\hat{z}_{\text{LNL}}(\psi) = \sum_{j \in \mathcal{N}_m(\psi)} a_j(\psi) \tilde{z}_j,$$

where

$$a_j(\psi) = \frac{\exp(-\|\psi - \psi_j\|^2/\tau)}{\sum_{k \in \mathcal{N}_m(\psi)} \exp(-\|\psi - \psi_k\|^2/\tau)}.$$

justify The weights are non-negative and satisfy

$$\sum_{j \in \mathcal{N}_m(\psi)} a_j(\psi) = 1.$$

Unlike the global linear lift, this construction allows the inverse approximation to vary across different regions of the embedding.

The scale parameter τ was fixed using the median squared distance between neighbouring observations in diffusion space, providing a characteristic local distance scale for the Gaussian weights **JUSTIFY THIS**. The neighbourhood size m was selected by comparing leave-one-out reconstruction error across a range of candidate values. The resulting parameter choice and reconstruction results are reported in Section 6.4.

Since the endpoints of each linear stress path correspond to known historical observations, their macroeconomic values are available directly. The lifted linear paths were therefore anchored to the exact observed start and stress states rather than using the reconstructed values produced by either lifting method at these two points.

To assess the ability of each lifting procedure to recover historical observations, leave-one-out reconstruction diagnostics were calculated. For each observation i , the corresponding macroeconomic state was removed from the training sample and the lifting procedure was constructed using the remaining $N - 1$ observations. The held-out diffusion coordinate ψ_i was then lifted to obtain $\hat{z}_{-i}$ and compared with the true state $\tilde{z}_i$. Reconstruction performance was summarised using the root mean squared error

$$\text{RMSE} = \sqrt{\frac{1}{ND} \sum_{i=1}^N \|\tilde{z}_i - \hat{z}_{-i}\|^2},$$

and the proportion of variance recovered,

$$R_{\text{lift}}^2 = 1 - \frac{\sum_{i=1}^N \|\tilde{z}_i - \hat{z}_{-i}\|^2}{\sum_{i=1}^N \|\tilde{z}_i - \bar{z}\|^2}.$$

These quantities were compared with a baseline reconstruction that assigns each held-out observation the mean of the remaining training observations **this is a very weak baseline, what else could you use? I now don't include the baseline score, but this is a part of the R^2 calculation, so you may want to explain it.** As the leave-one-out error is also used in selecting the neighbourhood size for LNL, these quantities are treated as cross-validated reconstruction diagnostics rather than as an entirely independent test set.

A second diagnostic was used to detect whether either lifting procedure mapped the interpolated linear paths unusually far from the historical macroeconomic cloud. For an interior lifted path point $\hat{z}_\ell$, define

$$d_{\text{NN}}(\hat{z}_\ell) = \min_{1 \leq j \leq N} \|\hat{z}_\ell - \tilde{z}_j\|.$$

The median and 95th percentile of these distances were calculated along each lifted linear path. The fixed historical endpoints were excluded from this diagnostic, since their nearest-neighbour distance is zero by construction. **Still might have endpoint effects? I think this is talking about if there is a point very close to the endpoint. Or it might be talking about the fact that a 95th percentile doesn't tell us about extremes**

To provide a reference scale, these path distances were compared with leave-one-out nearest-neighbour distances among the historical observations,

$$d_{\text{NN},-i}(\tilde{z}_i) = \min_{j \neq i} \|\tilde{z}_i - \tilde{z}_j\|.$$

The corresponding historical median and 95th percentile indicate the typical and upper-tail separation between observed macroeconomic states. The lifted-path diagnostic therefore tests whether the inverse mapping produces states lying unusually far from the empirical macroeconomic cloud. It is not interpreted as a direct measure of economic plausibility.

The lifting procedure used in the principal stress-path analysis was selected by considering both cross-validated reconstruction accuracy and the extent of extrapolation observed along the lifted linear paths. The numerical comparison between the Baker-style and local-neighbourhood lifts is presented in Section 6.4.

4.8 Nyström extension for out-of-sample embedding

Why even do this? New diffusion map is very quick to do computationally. Maybe it wouldn't be quick with lots of data. For oos testing we need this. For the HMM we train diffusion map on train set and then lift the rest into the sample for testing

The Nyström extension provides a way to embed a new macroeconomic observation into the previously learned diffusion space without recomputing the diffusion map or its eigendecomposition. The original embedding is therefore kept fixed, and the location of the new point is determined from its affinities to the historical observations.

Suppose that a new transformed macroeconomic observation

$$\tilde{z}_{\text{new}} \in \mathbb{R}^D$$

is to be embedded into the r -dimensional diffusion space. The extension uses the bandwidth ε , training degrees q_i , and diffusion-map eigenpairs $\{(\lambda_\ell, \phi_\ell)\}$ obtained from the original dataset $\tilde{Z}$. Here, ϕ_ℓ denotes the right eigenvector of the transition operator associated with λ_ℓ .

First, the affinity between the new point and each historical observation is calculated using the same Gaussian kernel as in the original diffusion map,

$$w_{\text{new},i} = \exp\left(-\frac{\|\tilde{z}_{\text{new}} - \tilde{z}_i\|^2}{\varepsilon}\right), \quad i = 1, \dots, N.$$

The corresponding kernel degree of the new observation is

$$q_{\text{new}} = \sum_{i=1}^N w_{\text{new},i}.$$

To remain consistent with the diffusion-map construction, the same α -normalisation is then applied,

$$w_{\text{new},i}^{(\alpha)} = \frac{w_{\text{new},i}}{q_{\text{new}}^\alpha q_i^\alpha}.$$

These affinities are row-normalised to produce a transition probability from the new point to each historical observation,

$$p_{\text{new},i} = \frac{w_{\text{new},i}^{(\alpha)}}{\sum_{j=1}^N w_{\text{new},j}^{(\alpha)}}.$$

The intuition behind the Nyström extension follows directly from the eigenvector equation for the transition matrix. For a historical observation i ,

$$\lambda_\ell \phi_\ell(i) = \sum_{j=1}^N P_{ij} \phi_\ell(j).$$

Thus, the value of the ℓ th eigenfunction at a point is related to a weighted average of its values at neighbouring observations. For the new point, the row $p_{\text{new},i}$ **What row?** plays the role of a new transition row. Requiring the same eigenvector relation to hold gives

$$\lambda_\ell \phi_\ell(\tilde{z}_{\text{new}}) = \sum_{i=1}^N p_{\text{new},i} \phi_\ell(i),$$

Have all these symbols already been defined? and hence

$$\phi_\ell(\tilde{z}_{\text{new}}) = \frac{1}{\lambda_\ell} \sum_{i=1}^N p_{\text{new},i} \phi_\ell(i).$$

The new observation therefore inherits its diffusion coordinates from the eigenfunction values of nearby historical observations, weighted according to its position relative to the learned data geometry. Importantly, the new point does not alter the existing eigenvectors or diffusion coordinates.

Using the same diffusion-time convention as in the original embedding, the out-of-sample diffusion coordinates are

$$\Psi(\tilde{z}_{\text{new}}) = (\lambda_1 \phi_1(\tilde{z}_{\text{new}}), \dots, \lambda_r \phi_r(\tilde{z}_{\text{new}})).$$

This procedure follows the Nyström extension principle used in [5], with the additional α -normalisation required to remain consistent with the diffusion-map construction used throughout this dissertation.

4.9 Gaussian HMM specification and evaluation

This bit is not well motivated

The diffusion-map analysis describes the geometry and historical support of macroeconomic states, but does not explicitly model their temporal evolution. As an additional analysis, a Gaussian Hidden Markov Model (HMM) was therefore fitted to the sequence of diffusion coordinates. The HMM is used to investigate whether observations in the learned embedding can be organised into persistent economic regimes and, subsequently, whether the constructed stress paths are consistent with the temporal structure learned from the historical data. **How? Maybe this will be explained in the background, or this can be a part of its own section**

Let $S_t \in \{1, \dots, K\}$ denote the latent state at time t . Conditional on $S_t = k$, the diffusion coordinates are assumed to follow a multivariate Gaussian distribution,

$$\Psi_t \mid S_t = k \sim \mathcal{N}(\mu_k, \Sigma_k),$$

where a full covariance matrix Σ_k is estimated separately for each state. The latent states evolve according to the first-order Markov transition probabilities

$$A_{ij} = \mathbb{P}(S_{t+1} = j \mid S_t = i).$$

The general HMM construction and sequence likelihood are described in the mathematical background.

The number of hidden states was selected by comparing models with

$$K \in \{2, 3, 4, 5, 6, 7\}.$$

Unclear For each K , the model was fitted from 100 different random initialisations in order to reduce sensitivity to local optima in the likelihood surface, and the fit attaining the largest log-likelihood was retained. For an r -dimensional diffusion embedding, the number of free parameters in the full-covariance Gaussian HMM is

$$p_K = (K - 1) + K(K - 1) + Kr + K \frac{r(r + 1)}{2}.$$

These terms correspond respectively to the initial-state distribution, transition matrix, state-dependent means and state-dependent covariance matrices.

Model complexity was first assessed using the Bayesian Information Criterion,

$$\text{BIC}_K = p_K \log N - 2 \log L_K,$$

Is this related to the Bayes factor, or the Occam factor? Not sure if this is important or just a point of interest where $\log L_K$ is the maximised log-likelihood of the observed diffusion-coordinate sequence and N is the number of observations. Lower BIC values are preferred, since an increase in model complexity must provide a sufficient improvement in likelihood to offset the additional parameters.

LLM pattern again Predictive fit was assessed separately using a chronological training–test split. The diffusion map was fitted using the first 70% of the observations, producing the training coordinates Ψ_{train} . The remaining 30% were embedded into this fixed diffusion space using the Nyström extension described in Section 4.8, giving Ψ_{test} . For each candidate K , an HMM was fitted to Ψ_{train} and the log-likelihood assigned to the held-out sequence Ψ_{test} was recorded. The BIC and out-of-sample likelihood **need not** favour the same model, and the resulting model-selection comparison is **therefore** discussed in the HMM results rather than imposing either criterion mechanically.

After selecting a specification, the inferred states were characterised using the corresponding macroeconomic observations. The most likely hidden-state sequence,

$$\hat{S}_{1:N},$$

was obtained from the fitted HMM, and the macroeconomic variables associated with each state were summarised to determine whether the statistical regimes admitted meaningful economic interpretations.

The inferred states were also compared with the NBER recession chronology. For each state k , define the binary indicator

$$I_t^{(k)} = \mathbb{1}\{\hat{S}_t = k\},$$

and let $R_t \in \{0, 1\}$ denote the NBER recession indicator. Agreement between $I_t^{(k)}$ and R_t was measured using Cohen's κ ,

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

where p_o is the observed proportion of agreement and p_e is the agreement expected from the marginal frequencies of the two binary series. Unlike raw classification accuracy, κ therefore accounts for agreement that would arise from the relative frequency of recession and non-recession observations. **this is very unclear**

Once a state exhibiting the clearest crisis or recession interpretation had been identified from the economic characteristics and external comparisons, its posterior probability was used for an additional lead-lag diagnostic. Let

$$\pi_t^{(c)} = \mathbb{P}(S_t = c \mid \Psi_{1:N})$$

denote the posterior probability of the selected crisis state. For offsets

$$\ell \in \{-6, \dots, 6\},$$

the correlation

$$C(\ell) = \text{Corr}(\pi_t^{(c)}, R_{t+\ell})$$

How? was calculated over the common time indices. Under this convention, $\ell > 0$ tests whether elevated crisis-state probabilities precede the corresponding NBER indicator, whereas $\ell < 0$ tests whether the HMM state tends to lag the NBER chronology. Since the posterior probabilities are conditioned on the complete observed sequence, this analysis is interpreted as a descriptive lead-lag diagnostic rather than an out-of-sample forecasting test.

The relationship between the HMM states and the empirical support of the diffusion embedding was also examined by grouping the diffusion-space density, or equivalently the density potential

$$V_t = -\log \rho_t,$$

Unclear according to the inferred state. This determines whether the statistical regimes occupy systematically different regions of the empirical density distribution. The economic interpretation of low-density observations is established separately in Section 6.2; consequently, the state-wise density comparison is treated as a coherence diagnostic rather than as an independent validation criterion.

Finally, we link this section back to the main theme of this dissertation by using our fitted HMM to provide a temporal consistency score for the stress paths constructed in Section 4.5. For a discretised path

$$\gamma = (\gamma_0, \dots, \gamma_L),$$

the fitted model assigns the sequence likelihood

$$p(\gamma_0, \dots, \gamma_L \mid \hat{\theta}),$$

where $\hat{\theta}$ denotes the estimated HMM parameters. Since the linear, ordinary graph and density-aware graph paths may contain different numbers of vertices, the comparison is based on the

log-likelihood per path point,

$$\mathcal{S}_{\text{HMM}}(\gamma) = \frac{1}{L+1} \log p(\gamma_0, \dots, \gamma_L \mid \hat{\theta}).$$

Larger values indicate that the sequence is more compatible with the regime distributions and transitions learned from the historical diffusion-coordinate series.

The vertices of a stress path do not necessarily **obvious? I think he means this is very obvious/not obvious...** represent equal monthly increments, particularly for graph paths. The HMM score is therefore used as a relative measure of consistency with the learned temporal structure rather than as the literal probability that the economy follows the scenario at monthly frequency. The model-selection results, economic state interpretations, NBER comparisons, density diagnostics and stress-path scores are reported in the dedicated HMM results section. **This whole section was unclear**

4.10 Principal component benchmark

To provide a conventional linear dimensionality-reduction benchmark for the proposed manifold-based stress paths, principal component analysis (PCA) was applied to the same quantile-standardised macroeconomic observations used to construct the diffusion map. The first $r = 4$ principal components were retained **is this a low number?**, matching the dimension of the diffusion embedding and thereby allowing the comparison to focus on the distinction between a linear PCA representation and the nonlinear diffusion embedding rather than on the number of retained coordinates.

Let

$$\Phi_{\text{PCA}} : \mathbb{R}^D \rightarrow \mathbb{R}^r$$

denote the fitted PCA transformation and let

$$y_i = \Phi_{\text{PCA}}(z_i)$$

be the PCA coordinates of observation z_i . For each of the start–stress event pairs introduced in Section 4.5, a linear path was constructed between the corresponding PCA coordinates,

$$\gamma_{\text{PCA}}(s) = (1-s)y_{\text{start}} + sy_{\text{stress}}, \quad s \in [0, 1].$$

The path was discretised using the same number of points as the corresponding density-aware graph path. This choice affects only the numerical representation of the continuous PCA interpolation and provides a comparable resolution for the subsequent diagnostics and plots.

Each interpolated point was mapped back to the quantile-standardised macroeconomic space using the inverse PCA transformation, **truncated PCA is not invertible. It's pseudo-invertible**

$$\hat{z}_{\text{PCA}}(s) = \Phi_{\text{PCA}}^{-1}(\gamma_{\text{PCA}}(s)).$$

Since only the first four principal components are retained, this inverse transformation reconstructs observations within the rank-4 PCA subspace and does not in general reproduce the original event endpoints exactly. The PCA endpoints were therefore left as their genuine rank-4 reconstructions rather than being manually replaced by the observed macroeconomic states. The resulting loss of information was quantified for each event using the mean endpoint reconstruction error

$$E_{\text{end}} = \frac{1}{2} (\|\hat{z}_{\text{PCA}}(0) - z_{\text{start}}\|_2 + \|\hat{z}_{\text{PCA}}(1) - z_{\text{stress}}\|_2).$$

This is inconsistent with what you did with LNL. Don't do this, or explain why you do. Explanation at the time was that I was comparing PCA to density-aware graph paths instead of PCA vs. LNL. I have actually changed this in the code/results. This provides a direct measure of how accurately the retained PCA representation captures the particular benign and stressed states used in the scenario analysis.

The location of the PCA path relative to the empirical macroeconomic data cloud was assessed using the nearest-neighbour diagnostic introduced in Section 4.7. For an interpolated PCA reconstruction $\hat{z}_{\text{PCA}}(s)$,

$$d_{\text{NN}}(\hat{z}_{\text{PCA}}(s)) = \min_j \|\hat{z}_{\text{PCA}}(s) - z_j\|_2.$$

Distances were calculated in quantile-standardised macroeconomic space and compared with the leave-one-out nearest-neighbour distances of the historical observations. The median, 95th percentile and maximum path distances were recorded. As in the lifting analysis, this diagnostic **is not** interpreted as a measure of economic plausibility; **rather**, it identifies whether the generated path passes through regions unusually distant from the observed macroeconomic sample.

For economic interpretation, the PCA reconstructions were subsequently mapped to the original units of the macroeconomic variables using the inverse fitted quantile transformation. Selected variables were then plotted along the PCA path and compared with the proposed density-aware graph path and the realised historical trajectory for each event. The plotted variables were chosen to reflect the macroeconomic quantities most strongly associated with the corresponding historical shock. Since the density-aware graph path consists entirely of observed diffusion-map nodes, its macroeconomic values are taken directly from the corresponding historical observations and require no inverse mapping. **This technically is an inverse mapping**

The PCA experiment therefore provides a method-level benchmark between a linear low-dimensional interpolation and the proposed nonlinear, density-aware graph construction. The endpoint reconstruction errors, nearest-neighbour diagnostics and macroeconomic trajectory comparisons are reported in Section 9.

4.11 Robustness design

In this section, we describe the robustness checks we performed to ensure the validity of our results. We hope **don't say this. We just test** to see that our qualitative result is stable: density-aware graph paths avoid low-density regions more effectively than straight-line interpolation. Additionally, we want **again, don't say this. This is a bias** to see a stable difference in the path diagnostics described in Section 4.6 between an ordinary and a density-aware graph path.

4.11.1 Sensitivity to core path parameters

The stress-path construction depends on several tuning parameters introduced in the preceding sections. Although the baseline values are selected using the diagnostics described earlier, it is important to establish that the principal conclusions are not specific to a single parameter configuration. Sensitivity analysis was therefore performed for the diffusion-map bandwidth ε , the number of nearest neighbours k used to construct the graph, and the density-penalty parameter β . In each experiment, one parameter was varied while the remaining parameters were held at their baseline values,

$$\varepsilon = 3, \quad k = 15, \quad \beta = 1.5.$$

The analysis was repeated for each of the four start–stress event pairs.

The bandwidth was varied over

$$\varepsilon \in \{1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5\}.$$

For each value of ε , the diffusion representation was reconstructed and the subsequent density estimation, graph construction and path calculation were repeated using the baseline values of k and β . The purpose of this experiment differs from the bandwidth diagnostics used to select ε in Section 4.3. Those diagnostics assess whether the diffusion operator itself is sufficiently well behaved, whereas the present analysis asks whether the resulting stress-path conclusions remain stable over a reasonable neighbourhood of the selected bandwidth.

Graph sensitivity was assessed using

$$k \in \{5, 10, 15, 20, 25, 30\}.$$

For each value of k , the symmetric nearest-neighbour graph was reconstructed on the fixed baseline diffusion embedding, after which both the ordinary and density-aware shortest paths were recalculated. This tests whether the conclusions depend strongly on the particular level of graph locality used to approximate movement through the sampled manifold.

For both ε and k , robustness was assessed by comparing the density-aware path directly with the corresponding ordinary graph path under the same parameter configuration. Let $M_{\min}$, M_{mean} , L_{diff} and D_{turn} denote respectively the minimum path density, arc-length-weighted mean density, latent path length and total turning angle defined in Section 4.6. For a generic metric M , define

$$\Delta M = M^{(\text{aware})} - M^{(\text{ordinary})}.$$

The four reported differences are therefore

$$\Delta M_{\min}, \quad \Delta M_{\text{mean}}, \quad \Delta L_{\text{diff}}, \quad \Delta D_{\text{turn}}.$$

Positive values of the first two quantities indicate that the density-aware path travels through regions of greater historical support than the ordinary graph path. By contrast, positive values of ΔL_{diff} and ΔD_{turn} indicate the geometric cost associated with this change, in the form of a longer or more strongly turning route. Expressing the results as differences isolates the effect of density awareness from changes to the underlying embedding or graph that affect both path types simultaneously.

The density penalty was varied over

$$\beta \in \{0, 0.25, 0.5, 0.75, 1, 1.5, 2\}.$$

Here the diffusion embedding, density field and k -nearest-neighbour graph were held fixed at their baseline specifications, and only the edge-cost penalty was changed. Since $\beta = 0$ reduces the density-aware objective to the ordinary graph shortest-path problem, a separate ordinary comparison is unnecessary in this experiment. Instead, the absolute values of

$$M_{\min}, \quad M_{\text{mean}}, \quad L_{\text{diff}}, \quad D_{\text{turn}}$$

were followed as β increased. This provides a direct view of the trade-off between avoiding regions of weak historical support and accepting additional path length or curvature.

No smooth dependence on these parameters is assumed. The stress paths are solutions to a discrete shortest-path problem, and changes in the embedding, graph connectivity or edge weights may cause the optimal route to switch between different sequences of observed states. Robustness is therefore assessed primarily through the persistence of the substantive comparison between density-aware and ordinary paths across the tested parameter ranges, rather than through smooth variation of the individual metrics. The corresponding sensitivity results are reported in Section 6.5.

4.11.2 Sensitivity to density estimation

The density-aware path construction depends on the estimate of historical support in diffusion space. We therefore test whether the principal empirical conclusion (that density-aware graph paths avoid regions of weak historical support relative to the linear and ordinary graph benchmarks) is robust to alternative definitions of the density field. In this experiment, the diffusion embedding, nearest-neighbour graph and density-penalty parameter are held fixed at their baseline specifications, while the density estimator used to construct the potential is varied.

The baseline estimator used throughout the main analysis is the graph-degree density introduced in Section 4.4. For a general point $x \in \mathbb{R}^r$ in diffusion space, its unnormalised density

estimate is

$$\hat{\rho}_G(x) = \sum_{j=1}^N \exp\left(-\frac{\|x - \psi_j\|^2}{h_{\text{dens}}}\right).$$

As previously, this is rescaled by the mean density at the observed diffusion coordinates,

$$\rho_G(x) = \frac{\hat{\rho}_G(x)}{N^{-1} \sum_{i=1}^N \hat{\rho}_G(\psi_i)}.$$

To assess sensitivity to this choice, we also consider kernel density estimators (KDEs). For a radial kernel K and bandwidth $b > 0$, the KDE at a general diffusion-space point x takes the form

$$\hat{\rho}_K(x) = \frac{1}{Nb^r} \sum_{j=1}^N K\left(\frac{\|x - \psi_j\|}{b}\right).$$

Three kernel forms are considered. The Gaussian kernel is

$$K_G(u) = \frac{1}{(2\pi)^{r/2}} \exp\left(-\frac{u^2}{2}\right),$$

the Epanechnikov kernel is

$$K_E(u) \propto \max\{1 - u^2, 0\},$$

and the exponential kernel is

$$K_X(u) \propto \exp(-u).$$

The Gaussian and exponential kernels have infinite support, whereas the Epanechnikov kernel assigns zero weight beyond the bandwidth. The exponential kernel also decays more slowly with distance than the Gaussian kernel. Since each estimated density is subsequently divided by its mean value over the observed diffusion coordinates, multiplicative kernel normalising constants do not affect the resulting normalised density field.

Before using alternative kernels, the Gaussian KDE provides a useful consistency check on the baseline graph-degree estimator. Substituting the Gaussian kernel gives the distance-dependent term

$$\exp\left(-\frac{\|x - \psi_j\|^2}{2b^2}\right).$$

This agrees exactly with the graph-degree kernel when

$$2b^2 = h_{\text{dens}},$$

or equivalently

$$b = \sqrt{\frac{h_{\text{dens}}}{2}}.$$

The remaining factors in the Gaussian KDE are constant with respect to x and disappear under the mean normalisation. The normalised Gaussian KDE should therefore reproduce the graph-degree density when this bandwidth transformation is used. This equivalence is used as an implementation consistency check rather than treating the Gaussian KDE as an additional robustness specification.

The robustness analysis consequently compares the baseline graph-degree estimator with Epanechnikov and exponential KDEs. For both alternative kernels, the bandwidth is fixed at

$$b = \sqrt{\frac{h_{\text{dens}}}{2}},$$

rather than being re-selected separately. This keeps the smoothing parameter fixed and allows the experiment to focus on the effect of changing the kernel form. The different kernels nev-

ertheless have different support and decay properties, so this choice does not imply that they induce identical effective smoothing.

For each density method m , the density is normalised using its own full-sample mean,

$$\rho^{(m)}(x) = \frac{\hat{\rho}^{(m)}(x)}{N^{-1} \sum_{i=1}^N \hat{\rho}^{(m)}(\psi_i)},$$

and the corresponding potential is constructed as

$$V^{(m)}(x) = -\log \left(\max\{\rho^{(m)}(x), \delta\} \right),$$

where δ is the same density floor introduced in Section 4.4. A new density-aware graph path is then obtained for each estimator by substituting $V^{(m)}$ into the density-weighted edge cost defined in Section 4.5. The ordinary graph path is unaffected by the density estimator and therefore remains fixed.

The geometry of the linear benchmark is likewise independent of the density estimator. To ensure that changes in its minimum evaluated density cannot arise from changes in discretisation, its number of interpolation points is fixed separately for each event using the baseline graph-degree specification,

$$n_{\text{lin}} = \left\lfloor \frac{n_{\text{ordinary}} + n_{\text{aware,baseline}}}{2} \right\rfloor$$

where n_{ordinary} and $n_{\text{aware,baseline}}$ are the numbers of vertices in the ordinary and baseline density-aware graph paths, respectively. The same linear grid is then used for all density estimators.

For each method, all three path types (linear, ordinary graph and density-aware graph) are evaluated under the density field associated with that method. At observed graph vertices, density is evaluated using the leave-one-out procedure from Section 4.6. In particular, for an observed node ψ_i , its own kernel contribution is removed while retaining the full-sample normalisation scale. For a KDE this corresponds to

$$\hat{\rho}_{-i}^{(m)}(\psi_i) = \frac{1}{Nb^r} \sum_{j \neq i} K_m \left(\frac{\|\psi_i - \psi_j\|}{b} \right).$$

Off-sample interior points of the linear path are evaluated against the full historical sample. The observed start and stress endpoints are excluded from the interior-density statistics.

The principal diagnostic is the minimum interior density,

$$M_{\min}^{(m)}(\gamma) = \min_{x \in \gamma^\circ} \rho_{\text{eval}}^{(m)}(x),$$

which is calculated for each event and path type under each density specification. Comparisons are made within a given estimator rather than between the absolute density values produced by different kernels. The relevant robustness question is whether the density-aware path continues to attain greater minimum historical support than the linear and ordinary graph paths when the definition of density is changed.

Finally, the density-aware graph paths generated by the different estimators are compared visually in diffusion space. This distinguishes robustness of the density ordering from robustness of the path geometry itself: two estimators may support the same qualitative conclusion while assigning different density values or selecting different sequences of historical states. The resulting density comparisons and path geometries are presented in Section 6.5.

4.11.3 HMM robustness

5 Numerical Validation on Toy Model

5.1 Purpose of the validation experiments

5.2 Benchmark manifolds

5.3 Bandwidth and spectral validation

5.4 Recovery of manifold geometry

6 Macroeconomic Manifold and Stress Paths

This is the main results section of the paper

6.1 Empirical embedding results

The diffusion-map construction was applied to the quantile-transformed macroeconomic observations using the density normalisation $\alpha = 1$. The bandwidth and embedding dimension were selected using the diagnostics described in Section 4.3. Since the purpose of this stage was to obtain a suitable geometric representation for the subsequent path analysis, rather than to interpret every individual diffusion coordinate, the results are reported only briefly.

Bandwidth selection. The kernel-mass scaling diagnostic suggested a bandwidth of approximately $\varepsilon = 1.13$. This value was rejected because the corresponding minimum kernel degree was close to one. As the diagonal affinity satisfies $W_{ii} = 1$, this indicated that some observations were almost isolated from the remainder of the sample. The leading diffusion coordinates at small bandwidths were also largely flat, with occasional sharp spikes, rather than displaying stable variation across the sample. Include a plot of this maybe

A bandwidth of

$$\varepsilon = 3$$

was selected instead. At this value, the minimum degree increased to approximately 19.35, and the degree distribution and leading eigenspectrum were stable relative to nearby bandwidths. Increasing the bandwidth further did not materially improve the leading coordinates and would have made the kernel progressively less local. The selected value was therefore the smallest bandwidth that provided a sufficiently connected and spectrally stable diffusion operator.

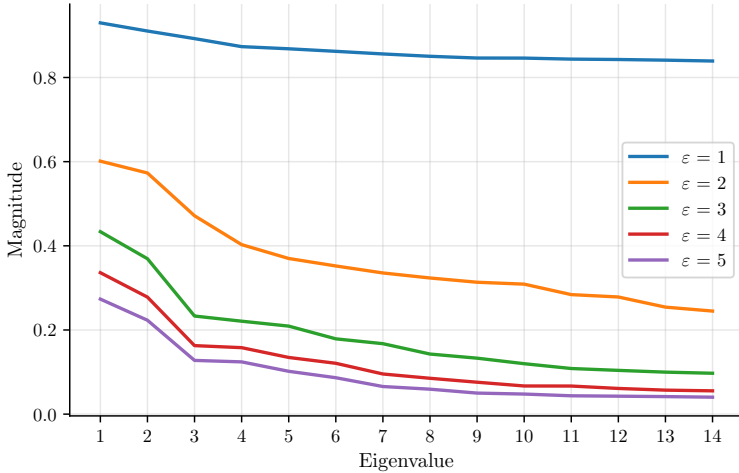


Figure 1: Spectrum of the transition matrix for different bandwidths.

ε	Min Degree	Degree CV
1	1.20	0.47
2	5.54	0.37
3	19.35	0.28
4	41.77	0.23
5	69.34	0.19

Table 1: Minimum degree and degree CV across bandwidths.

Embedding dimension.

- spectrum suggests $r=2$;
- visualisations and interpretation make $r=3$ and $r=4$ plausible;
- near-collision and reconstruction diagnostics favour $r=4$;
- correlation analysis shows that the third and fourth coordinate captures a distinct macroeconomic axis - we can interpret these;

- $r=5$ gives little additional improvement from near-collision statistics

Change this section a bit. Add largest spectral gap suggesting $r = 2$. 3d plots suggest $r = 3$, although hard to check for 4d probably; can give this a go plotting the 2,3,4 coordinates and seeing if there is a separation in two different directions like CPI vs UNRATE for coordinates 1,2,3. LNL RMSE and % variance recovered suggest $r = 4$. Near-collision comparison suggests $r = 4$. We choose $r = 4$ because ultimately we are interested in the reconstruction back to macro variables. The leading eigenspectrum did not, by itself, provide an unambiguous cut-off for the embedding dimension. The incremental information supplied by additional coordinates was therefore assessed using the near-collision diagnostic. For each candidate dimension r , pairs of observations that were close in the first r diffusion coordinates (closest 1%) were identified, subject to a twelve-month temporal exclusion window. The separation of these pairs in the the higher-dimensional space was then measured and percentiles were taken.

Table 2 reports the results. The addition of coordinate $r = 3$ and $r = 4$ both produced meaningful separation of observations that appeared close in the lower-dimensional representation. By contrast, coordinate $r = 5$ provided substantially weaker separation, indicating that the incremental geometric information beyond dimension $r = 4$ was limited. The final embedding dimension was therefore fixed at

$$r = 4.$$

Table 2: Percent of the near-collision pairs that are no longer in the closest $x\%$ of pairs in the higher-dimensional embedding. Near-collisions are pairs are the top 1% of closest nodes in the lower-dimensional embedding after applying a twelve-month temporal exclusion window.

Comparison	Near-collision pairs	% Exit 1%	% Exit 5%	% Exit 10%
2D $\rightarrow$ 3D	2669	58.0	31.1	18.6
3D $\rightarrow$ 4D	2669	49.5	20.8	10.1
4D $\rightarrow$ 5D	2669	23.1	1.95	0.26

In Table 2, we can see that adding an extra dimension can add lots of separation. For example, roughly 496 of the near-collision pairs in $r = 2$ are no longer in the top 10% closest pairs in $r = 3$; this clearly shows that there is a considerable amount of new information gained by adding a 3rd coordinate. The results are similar for adding the 4th coordinate; however, there is a dramatic drop-off in the gained separation of near-collision nodes when adding the 5th coordinate.

Structure of the embedding. Figure 2 displays the resulting diffusion coordinates. Observations associated with NBER recessions were visibly displaced from the main concentration of observations in the leading projections. This separation was already present in the first few coordinates, indicating that recessionary periods represent comparatively unusual locations in the learned macroeconomic state space rather than being distinguished only by higher-order coordinates.

The embedding also exhibited systematic variation with NBER labels, inflation rates, and industrial production levels. These patterns support the interpretation of the diffusion coordinates as describing economically meaningful variation in the joint state of the variables. The coordinates themselves are not assigned direct economic labels, however, and are used primarily as a geometric state representation in the subsequent graph and stress-path analysis.

Overall, the selected diffusion map provided a connected and stable low-dimensional representation of the transformed macroeconomic data. This embedding is used as the state space for the graph-based stress paths constructed in the following section.

6.2 Embedding density results

When generating the k NN graph from the diffusion coordinates using the metrics described in Section 4.4, the results over many values of k were very stable. With no clear winner, we chose

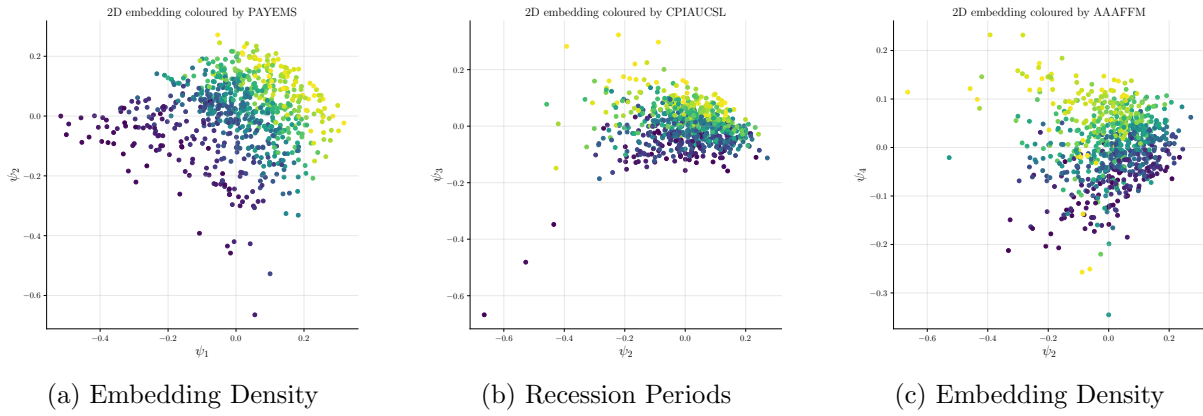


Figure 2: Diffusion-map representation of the quantile-transformed macroeconomic observations using $\alpha = 1$, $\varepsilon = 3$. Plotted against different coordinates and coloured by different variables to show the different structures found by different coordinates.

$k = 15$ as our final value. The results are displayed in Table 1.

Table 3: Network Metrics by k

k	Mean degree	Graph density	Jump ratio (95th percentile)
5	6.5734	0.0089	2.2838
10	12.8668	0.0173	2.3448
15	19.3674	0.0261	2.3373
20	25.8089	0.0348	2.3475
30	38.7645	0.0522	2.3288

For the bandwidth of the graph-degree density kernel, we chose it to be one-tenth of the median squared distance between diffusion coordinates. Choosing a bandwidth that was too small resulted in many points being classified as low-density, whereas choosing one that was too large meant the density was very similar across a wide range of points. The resulting density of points was similar to nearby choices of this bandwidth. The density floor δ was chosen to be 10^{-6} ; however, with the chosen bandwidth all the resulting densities were larger than this value, so it did not affect downstream results. **I think that this can all just go in the methodology section. These are not really results, and there is not much to say about these choices. We just had to make them.**

The density of a point in diffusion gives us a measure of how close to other historical diffusion points. We would like to understand whether the density has any meaning outside of diffusion space. It is hypothesised that a low density point may correspond to a state of the economy which is an outlier. To test this, we plot the density penalty (V) of points over time, and we shade in regions of recessions, as defined by the National Bureau of Economic Research. **Why do we plot penalty, not just density? Because we work with the density penalty in the density-aware cost, but why??** The results are shown in Figure 3.

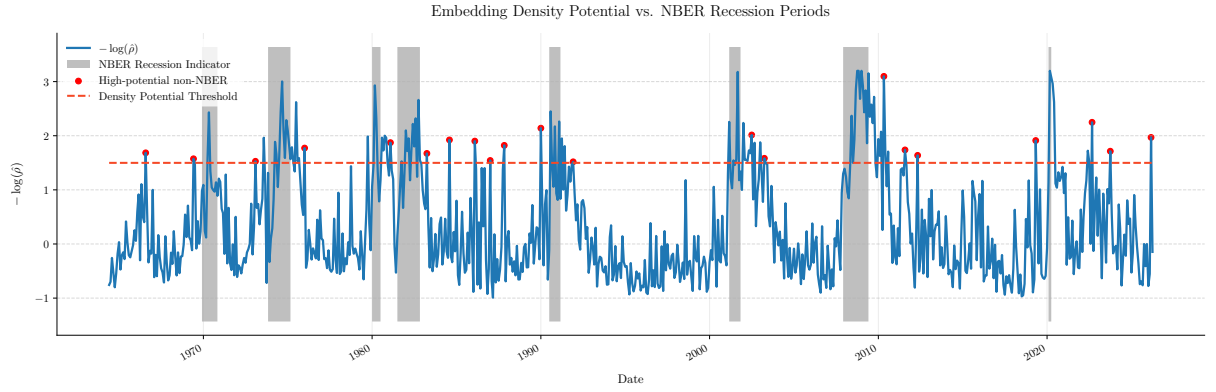


Figure 3: Plot of the density penalty of diffusion coordinates (V) over time with NBER recession dates shaded grey. We also highlight points with high density penalties, but not in NBER recession dates.

We can see in Figure 3 that low-density points are well captured by the NBER recession points. During every recession, there is a local minimum density value. However, we can identify points which are not classified as recessions. We classify these points by taking the median value of the density penalty in the NBER states. We then find which dates correspond to density penalty values greater than or equal to this value. We ignore points within 4 months either side of a recession because they can be classified as pre-recession or post-recession points, which are historical outliers **are they? why? source?**. Among the identified points, highlighted in red in Figure 3, we find that all correspond to a time of economic outlier; for example, the most recent red point corresponds to the start of the U.S.–Iran war, where the price of oil rose to \$100 per barrel **Citation? This last paragraph feels a bit non-academic**

6.3 Stress-path results

The events and dates in Table 4 were selected based on both their macroeconomic variables and news articles indicating important historical dates. For our benign states, we inspected several macroeconomic variables to ensure that the economy was not in a strange state. For the stress points, we compared the macroeconomic variables and various news sources to identify regions of genuine economic stress. For the investigated five pairs, however, only four were retained due to one pair being too close in diffusion space. This choice was made before further testing commenced to avoid bias. The chosen endpoints are displayed in Table 4, and an in-depth discussion of the dates is included in Appendix A.

Table 4: Macroeconomic Regimes and Event Analysis

Regime / Macro Event	Start Date	End Date
Benign Expansion to Global Financial Crisis (GFC)	2006-03-01	2008-10-01
Benign Expansion to COVID-19 Shock	2019-07-01	2020-04-01
Low-Inflation to Monetary Tightening	2019-04-01	2022-04-01
Pre-Crisis to Early 1980s Recession Trough	1977-01-01	1982-07-01

In Figure 4, we plot the resulting paths for each scenario on a 2D projection of our embedding.

While this projection gives us some idea of difference between the lines, we can plot the density along each of the curves to get a better understanding of the lines themselves. We see that in Figure 5.

Real results from Phase 6, show trade off between density and length. Show tables/figures of results for each event for each path type. GFC and COVID have low linear density because they are very far outside of main cloud, the other two have more reasonable min density on

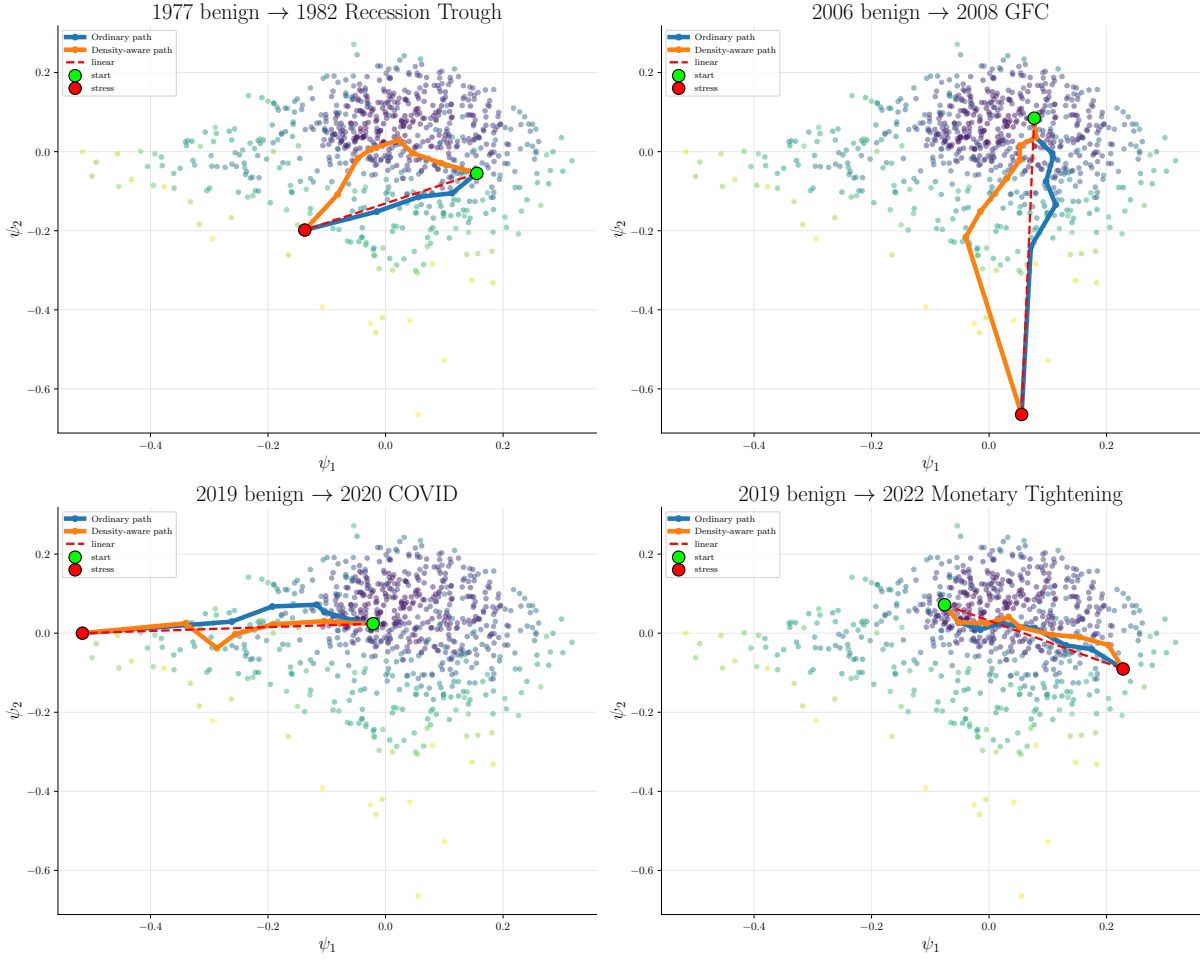


Figure 4: Caption

lin path. Total turning angle is just used to show that the path is turning more to reach these points of high density.

In Table 5 we can clearly see that the density aware paths have the highest minimum density for each of the events. Similarly, the mean density for every event is the highest for the density-aware paths. There is an interesting result for the mean density metric, however. For the 1977 → 1982 recession event, the mean density for the linear path is actually higher than the mean density for the ordinary graph path **EXPLAIN THIS - can use the maths behind the minimisation argument**. By construction, the linear paths must always have the shortest total path length, as it is a straight line through diffusion space. We do see the an increase in the distance of the paths from the ordinary graph path to the density-aware graph path. This reflects the path able to sacrifice some length to minimise the density penalty in the edge cost function

$$w_{ij}^{(\beta)} = \|\psi_i - \psi_j\|^2 \exp\left(\beta \frac{V_i + V_j}{2}\right).$$

We can see that the total angle turned also increases from the ordinary graph path to the density-aware graph path, which reflects the path taking more jagged routes through the diffusion coordinates to reach the higher density regions. Both the length of the path and the total angle turned by the path don't tell us anything economically important about the path; they are useful metrics to see that our density cost function is working. For example, in the 2019 → 2022 tightening scenario, it is hard for us to tell that the path is longer or more jagged while looking at a 2D projection, but the length and angle metrics along with the density metrics allow us to see that the path is fundamentally changing route because of the density penalty.

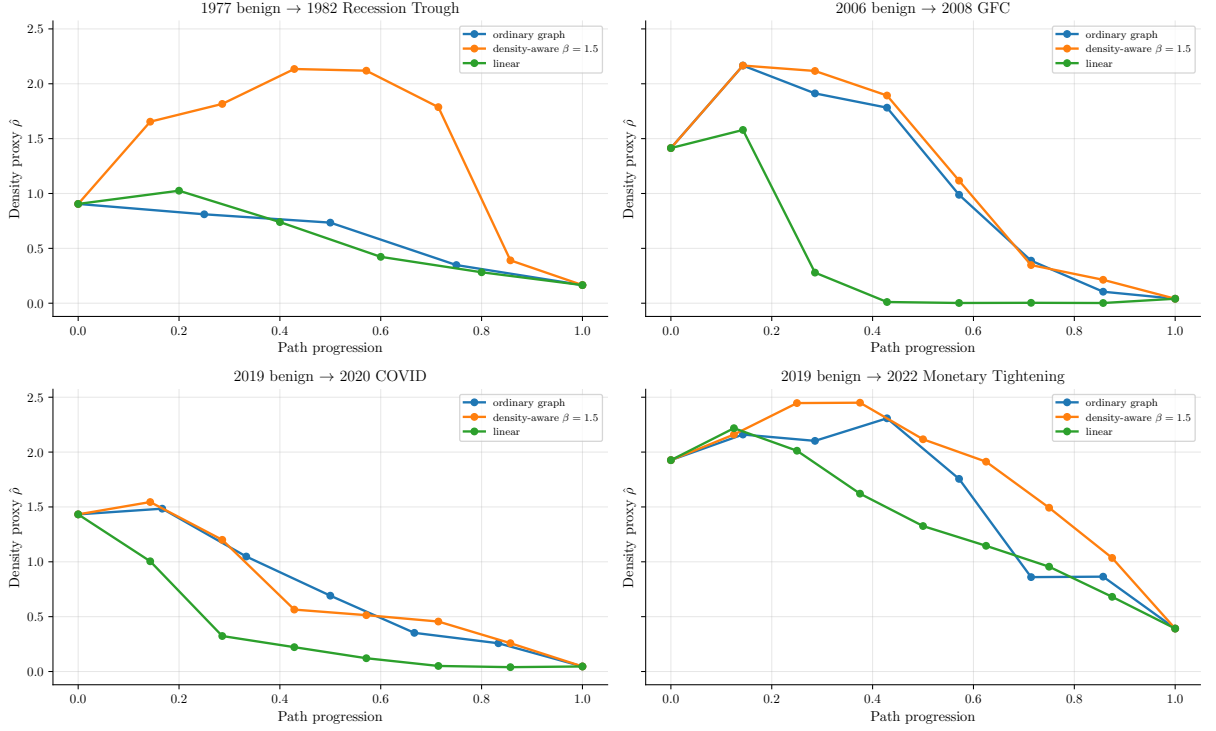


Figure 5: Caption

Table 5: Stress-path evaluation across the four historical scenarios. Higher values of $M_{\min}$ and M_{mean} indicate greater historical support. L_{diff} is measured in diffusion-coordinate space and D_{turn} is the total turning angle.

Scenario	Path	$M_{\min}$	M_{mean}	L_{diff}	D_{turn}
1977–1982 recession	Linear	0.2821	0.6009	0.3393	0.0000
	Ordinary graph path	0.3070	0.5056	0.3827	2.3796
	Density-aware	0.3499	1.1990	0.5026	4.5465
2006–2008 GFC	Linear	0.0019	0.3720	1.0611	0.0000
	Ordinary graph path	0.0640	0.4187	1.1114	4.0343
	Density-aware	0.1725	0.4860	1.1473	5.1949
2019–2020 COVID-19	Linear	0.0398	0.3558	0.6530	0.0000
	Ordinary graph path	0.2168	0.4409	0.7024	3.8314
	Density-aware	0.2174	0.4612	0.7410	4.3808
2019–2022 tightening	Linear	0.7236	1.3872	0.3549	0.0000
	Ordinary graph path	0.8196	1.4472	0.4140	4.9406
	Density-aware	0.9945	1.6850	0.4350	5.7592

6.4 Lifted-path results

In [1], they used a linear lifting operator to lift the diffusion coordinates to macroeconomic variables. The point of using a new method is to try to improve on the linear interpolation method by giving even naive linear paths more historical support. **The whole point of the diss is that we are constrained to use diffusion coordinates and now we want to find a good way to model things while using them. So even the linear interpolations we want to improve.** For the LNL method, we chose $m = 16$ neighbours which minimised the RMSE of the method for the chosen range of choices.

We results of the reconstruction tests described in Section 4.7 show that local-neighbour lifting is marginally better at reconstructing the sample of macroeconomic variables, with an R^2 score of 0.458, compared to a score of 0.436 for the Baker-style method. This means that, compared to simply projecting every diffusion coordinate to the mean of the remaining macroeconomic variables, LNL reconstructs the sample marginally better. Results are displayed in Table 6.

Table 6: Model Performance Metrics

Method	RMSE	R^2
Baker linear lift	1.456581	0.435582
Local-neighbour lift	1.426920	0.458335

We also test the nearest neighbour statistics of paths for each event for both methods. This test is to analyse the reconstructions of the paths and compare to the historical data cloud.

Table 7: Nearest-neighbour diagnostics for the lifted linear paths. Distances are measured in the quantile-standardised macroeconomic space. The ratio R_{95} compares the 95th percentile of the path distances with the 95th percentile of the leave-one-out nearest-neighbour distances among the historical observations. Values of $R_{95} > 1$ indicate that the lifted path contains points farther from the historical data cloud than is typical at the corresponding historical threshold.

Scenario	Lift	Median d_{NN}	R_{95}
1977–1982 recession	Global linear	1.201	0.693
	LNL	1.132	0.643
2006–2008 GFC	Global linear	1.589	1.665
	LNL	1.103	0.676
2019–2020 COVID-19	Global linear	1.200	0.976
	LNL	1.103	0.669
2019–2022 tightening	Global linear	1.189	0.683
	LNL	1.184	0.689

In this test we can see that the local-neighbour lift performs better than the Baker-style lift on every metric (apart from Median d_{NN} for 2019 $\rightarrow$ 2022 Monetary tightening). All of the results suggest that both of these methods work reasonable well because the median distance from path points to the observation data cloud is less than the average distance between points in the data cloud. This is not the case for the Global Financial Crisis path with the Baker-style lift, where the 95% quantile of the distance to the closest neighbour is roughly 3 compared to the historic 95% quantile of approximately 1.8. This shows that the Baker-style lifting may not be suitable for all stress scenarios, because it is sending points in diffusion space to points very far away from points in macro space. The results show that even though we are using a naïve approach of linear interpolation, we can use a modified approach to lifting unseen points to macro space, which seem to pull points closer to the observed data cloud. This may prove useful

in future work. For the rest of the dissertation, when lifting linear paths to macroeconomic variables, we will use LNL.

The aim of this dissertation is to generate paths along the learned geometry that remain historically supported, not directly temporally reasonable. This is worth baring in mind when looking at the following figures, where we plot a selection of macroeconomic variables along generated stress paths. We also plot the real path of the macroeconomic variable to compare. While we don't directly expect the generated path to follow the real trajectory, if one does then we can interpret that as the path following a historically supported path.

There will now be a lot more discussion about these figures. Explain most of them propably. We're not necessarily looking for a match, like I've mentioned. 1982 UNRATE looks bad, the variations are massive, linear path seems to follow better.

We can see in Figure 7a that the denstiy-aware graph path not only follows only historically observed nodes, but the order of the nodes in the paths follows a historically observed order. We can also see that the linear path does not follow the shape of the historical path, this doesn't mean that the linear path 'falls off the manifold', but it means that the path may represent a combination of variables which may correspond to extreme economic regimes **HOW? DOES THIS REALLY MAKE SENSE?**

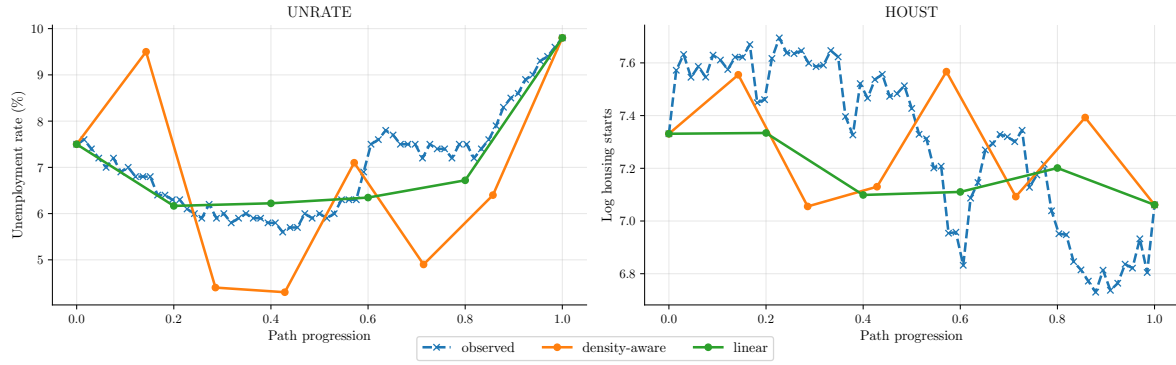
We can also compare to PCA paths. **HOW DO WE CORRECTLY CHOOSE THE RIGHT DIMENSION FOR PCA?** Right now I choose $r=4$ because that's the same as for diffusion map. We see in Figure 8 that the PCA paths are very straight. This is expected and is quite a rudamentary comparison. The rest of the figures are in Appendix B. An exploratory PCA implementation indicated that graph-constrained paths may also produce coherent routes under a linear embedding. This suggests that part of the improvement over direct interpolation arises from graph-based support constraints, rather than solely from the nonlinear diffusion representation. A systematic comparison of alternative representations is left for future work.

Interpret these paths. Maybe here include the PCA comparison

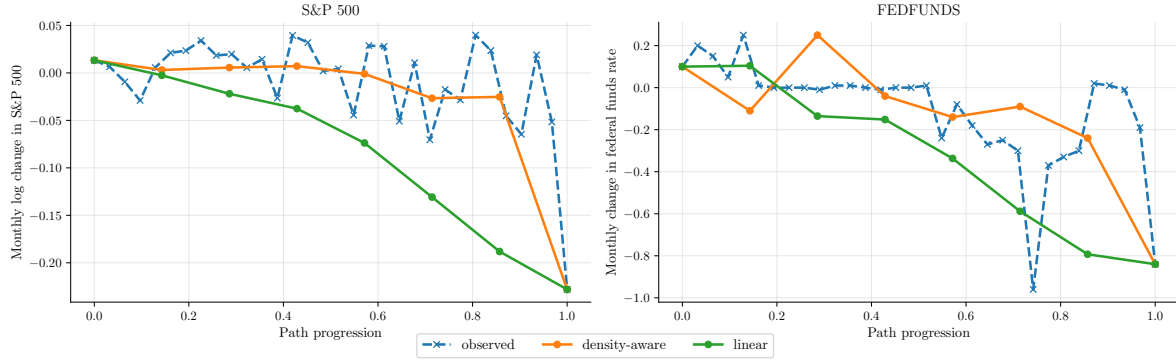
6.5 Robustness of parameters

In this section we describe what happens to our model when we change certain parameters. Firstly, we vary the sparsity penalty term β and investigate what happens to the four key metrics for our density-aware graph paths: minimum density, weighted mean density, diffusion length, and total turning angle. We see in Figure 9 that for minimum density, mean density, and diffusion length, there is a monotonioic increase in these values as β increases. The increase is not drastic and for some levels of β , the metrics don't move. This reflects the fact that an optimal path for one sparsity penalty level may be the same path for another. The results for the total turning angle metric are similar, with a general increase as β increases. This affect has been explained in Section 6.3 **MAYBE NOT THE RIGHT REFERENCE**. The non-monotonicity of this metric does not raise any concerns as it simply reflects a smoother route being better for the optimisation for certain levels of β **THIS MIGHT NEED TO BE EXPLAINED MORE, IF YOU EXPLAIN IT WELL ENOUGH BEFORE, WHY IT MAKES SENSE TO GO UP, YOU SHOULD BE ABLE TO EXPLAIN WITHOUT CONTRADICTION WHY IT MAY GO DOWN TOO. IT DOESN'T ACTUALLY SEEM THAT INTUATIVE**

We then vary the bandwidth parameter ε in the diffusion map construction and the number of neighbours k in the k NN graph construction seperately. For every parameter choice, for every tested scenario, we find the minimum density of our density-aware graph path is greater than or equal to the minimum density of the ordinary graph path. This suggests that our density-aware cost function is robust. We do, however, find that for $\varepsilon = 2$, the weighted mean density of our density-aware path is lower than that of the ordinary graph path. **THIS CA BE EXPLAINED AGAIN USING THE FACT THAT THE MINIMISOR DOESN'T ACTUALLY CARE ABOUT THE MEAN**. We can also see a stable and consistent picture with the latent length and total turning angle metrics, with almost all selected parameters showing the diffusion distance and total turning angle metrics are higher for the density-aware paths. This is not the case for $k = 20$ for the turning angle metric, **This should have been explained in first paragraph, so reference**

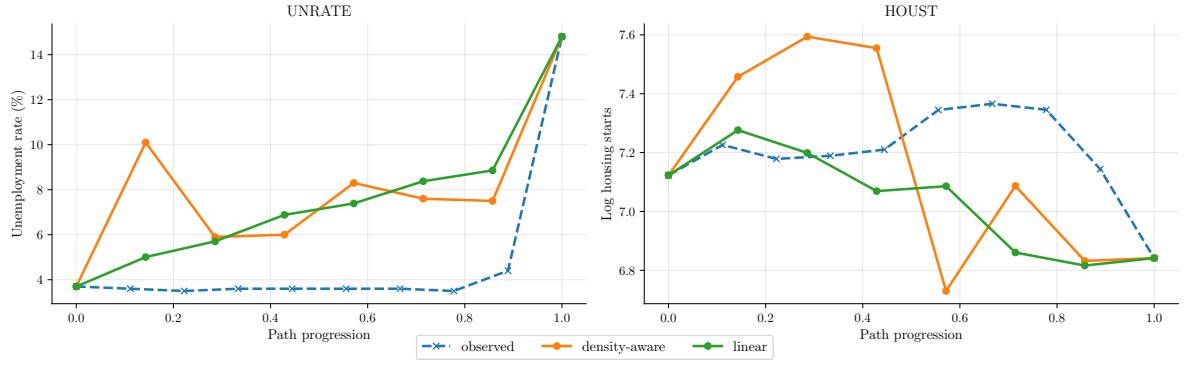


(a) Transition from the benign state in 1977 to the 1982 recession trough.

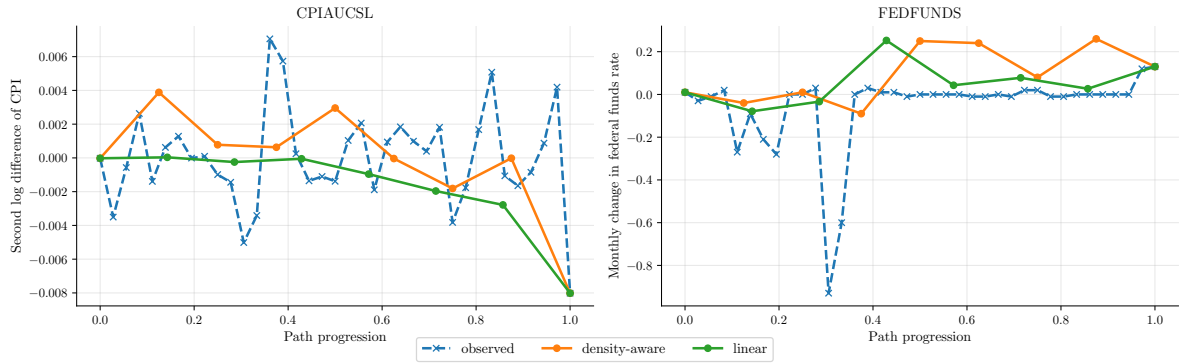


(b) Transition from the benign state in 2006 to the Global Financial Crisis in 2008.

Figure 6: Lifted macroeconomic trajectories for the early-1980s recession and the Global Financial Crisis. The graph path consists of observed transformed macroeconomic states, whereas intermediate points on the linear diffusion-coordinate path are reconstructed using local neighbourhood lifting.



(a) Transition from the benign state in 2019 to the COVID-19 disruption in 2020.



(b) Transition from the benign state in 2019 to the monetary-tightening episode in 2022.

Figure 7: Lifted macroeconomic trajectories for the COVID-19 disruption and the 2022 monetary-tightening episode. The graph path consists of observed transformed macroeconomic states, whereas intermediate points on the linear diffusion-coordinate path are reconstructed using local neighbourhood lifting.

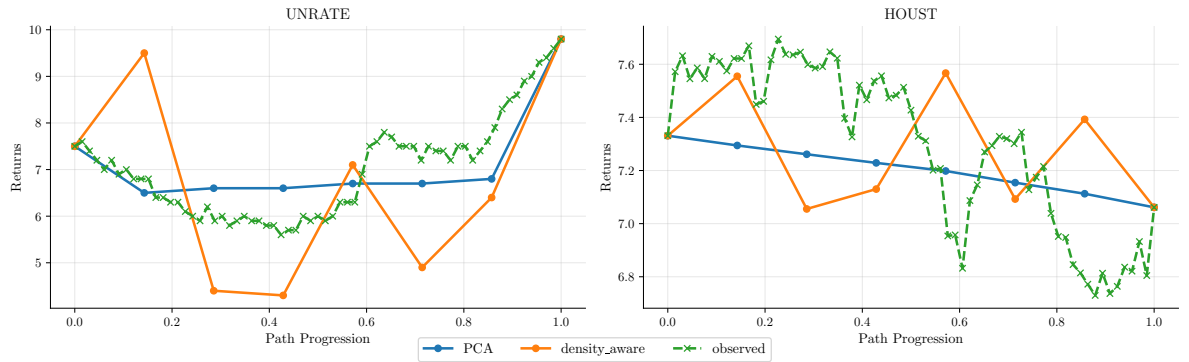


Figure 8: Caption

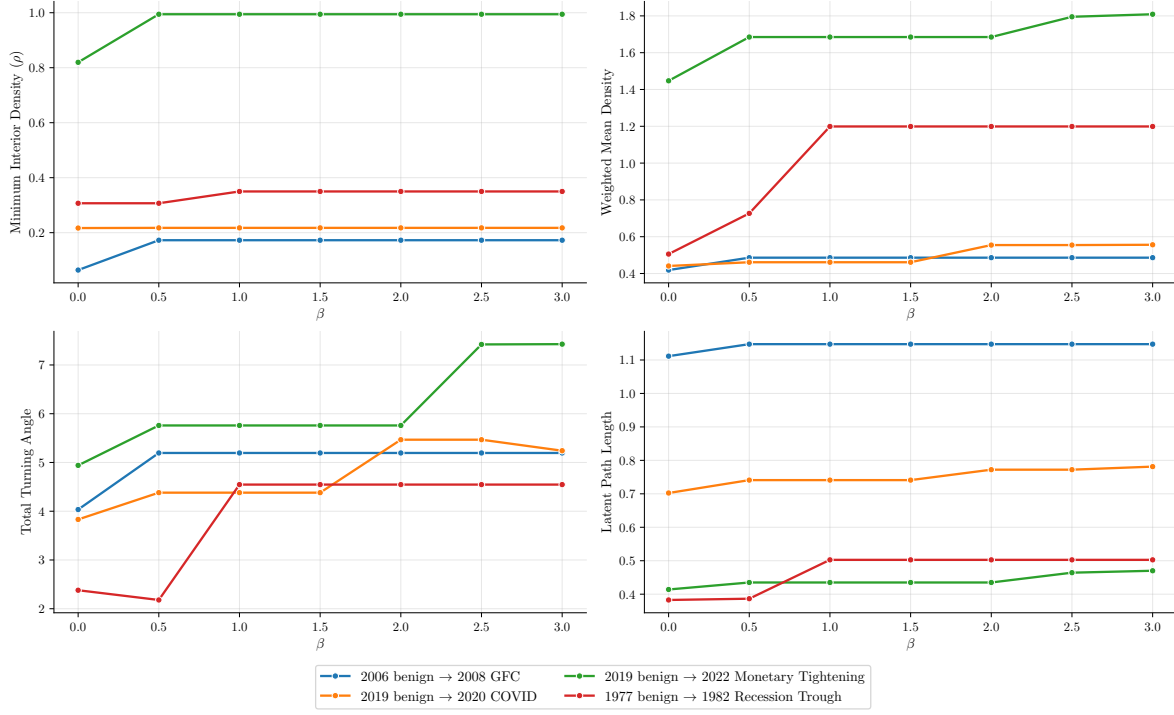


Figure 9: Caption

that.

Finally, we explore what happens to our main result (density-aware paths avoid sparse regions while linear paths do not), when we change the density estimation function. We compare our results to using kernel density estimates with 3 different kernels. First, we use the Gaussian kernel where we select the bandwidth to be such that the resulting densities should be identical. This was simply to validate our density calculations and we found the desired result. Secondly we used the exponential kernel and the Epichinikov kernel to estimate density. The exponential kernel gives more weight to further away points, whereas the Epichinikov kernel has finite support and assigns to weight to points for which $|u| > 1$. Using the new density estimates, we calculate new density-aware graph paths between endpoints. We also must re-evaluate the density of points along the linear paths and ordinary graph paths, even though they do not change under the new density estimates. We compare the minimum density of the linear path, ordinary path, and density-aware path in Figure 10.

We can see in Figure 10 that for each kernel, and for each event, the density-aware path has the highest minimum density. In almost all cases the linear path has the lowest minimum density, apart from for the Epanechnikov kernel for the 2022 Monetary tightening event. This even has already raised a few strange results, so it is not surprising we would see another result like this. As mentioned before, both the start and the stress state lie in a high density region, so a linear path through this region is not automatically a bad path to choose. This is reflected in the other two kernels for this scenario, where the minimum density of all three paths are quite similar. **EXPLAIN MORE ABOUT EPACHNIKOV DIFFERENCE.**

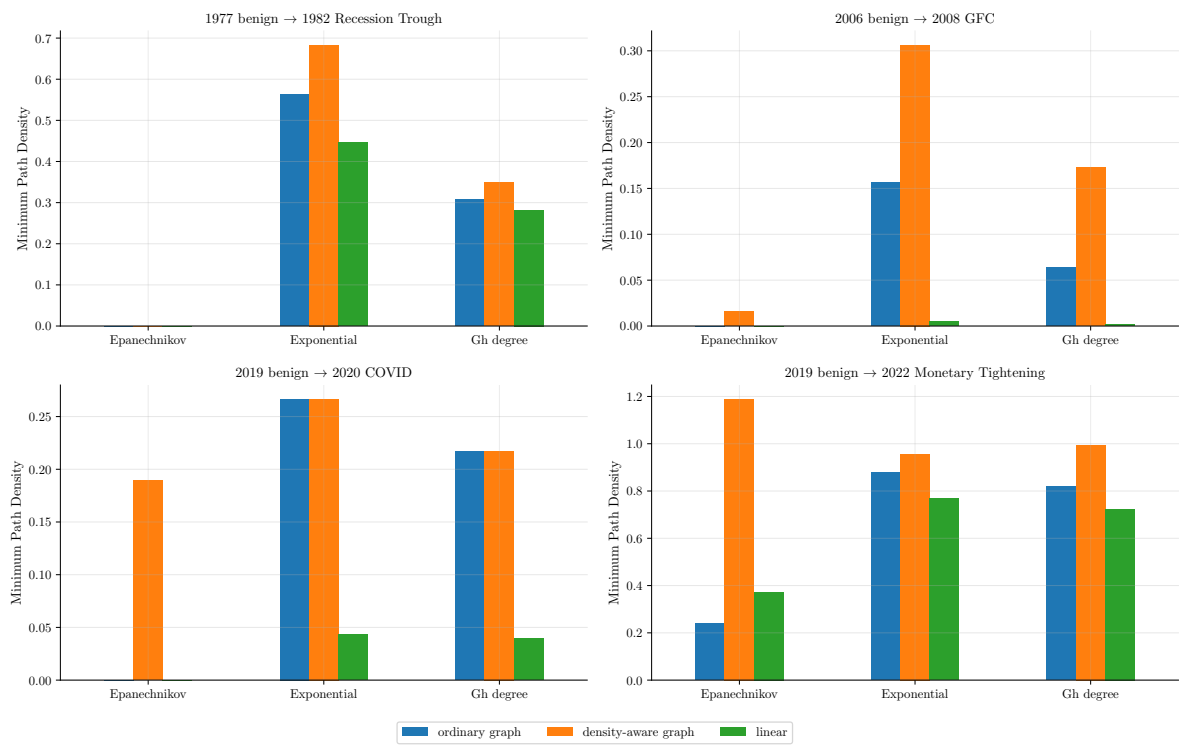


Figure 10: Caption

7 Markov bridges

7.1 Markov bridge motivation and background

A natural extension is being able to find a distribution of paths. In modelling, you rarely want to return a deterministic answer; rather, you would like to see a distribution of scenarios with corresponding probabilities. Therefore, the question is, can we generate a distribution of stress paths between two endpoints which follow the learned geometry? This section introduces a method which can be used to generate a series of stress trajectories which travel along the learned geometry. We also introduce a density-aware methodology which is analogous to the main part of this dissertation. The aim of this section is twofold: to implement a way to generate sample paths which travel along the learned manifold and avoid weakly historically supported regions, and to strengthen our findings of the density-aware graph paths, by seeing whether they represent a path along the whole graph random walk by using only the leading eigenvalues of the transition matrix.

In our construction of the diffusion map, we create a transition matrix P , whose elements P_{ij} correspond to the probability of node i jumping to node j in a single step of a graph random walk. We can extend and find that $(P^k)_{ij}$ is the probability that node i jumps to node j in k graph random walk steps. This probability matrix contains information about the underlying geometry because of the bandwidth parameter we chose to uncover the local connections.

A Markov bridge is constructed by conditioning a Markov process to start at a specific point and end at a specific point over a certain number of random walk steps. We can then get a series of random variables

$$X_0, X_1, \dots, X_K,$$

where K is the number of steps in the Markov bridge, and X_0 and X_K are the conditioned start and end points, respectively.

These paths then represent possible paths through observed points from the start state to the stress state. The jumps are not bounded by the k nearest neighbours like in the density aware graph paths **what are the implications of this? This means things are jumping off the manifold in our jump, like setting k too high in graph. The long jumps have low probabilities, and could happen in real life potentially. Worth testing with the computed length of jumps for consecutive nodes in our sample?** The aim of our work was to build paths between endpoints, through highly supported regions. The Markov paths only have a notion of local connectivity, not density. **This is quite correlated I would assume, the high density points have highly connected neighbours e.g.** Then instead of taking simple Markov bridge paths, we introduce a density-aware Markov bridge path via

$$\bar{P}_{ij} \propto P_{ij} \exp(-\alpha V_j),$$

where V_j is the density potential introduced in Section 2.3 and α is a strength of sparsity avoidance. This has the effect of reducing the probability that node i jumps to node j if node j is a low-density node.

Here you can maybe explain more background on Markov bridges

7.2 Markov bridge methodology

To sample from a Markov bridge, we don't need to generate the entire step-dependent transition matrix Q_k ; instead, we can introduce

$$h_k(i) := \Pr(X_K = S | X_k = i) = (P^{K-1})_{iS},$$

which tells us the probability that we will reach the required stress state, given that we are currently at node i . These vectors allow us to condition our transitions via the following

$$\Pr(X_{k+1} = j | X_k = i, X_K = S) = P_{ij} \frac{h_k(j)}{h_{k-1}(i)}.$$

We can see that, in our Markov bridge, the probability of going from node i to node j is increased if the probability of going from node j to node S in the remaining steps is high (relative to the other nodes). To generate a path, we start at a specific node i and use this formula to figure out the probability that we transition to the next node j . We can do this until we reach S and repeat for however many simulations we want.

We build the vectors h via backwards recursion with the end condition that

$$h_K(i) = \begin{cases} 1 & \text{if } i = S \\ 0 & \text{if } i \neq S. \end{cases}$$

This condition says that the last node in the chain must be the stressed node **Why does it say that? Because of what we defined above with the $P \frac{h_1}{h_0}$ thing.** We work backwards from this to find h_{K-1} . We remind ourselves that

$$h_{K-1}(i) = \Pr(X_K = S | X_{K-1} = i),$$

in words: the probability that we end up at the stressed node, given that we have got one step to get there. This simply corresponds to the probability of jumping from any node to S which is $P_{\cdot,S}$. This is equivalent to Ph_K , therefore

$$h_{K-1} = Ph_K,$$

thus the start of the recurrence relation. For illustration purposes, we explain how $h_{K-2} = Ph_{K-1}$. We want to know the probability of landing at the stressed node given that we have two steps of our random walk remaining. We know that h_{K-1} tells us the probability of finding a path from any node to the stressed node in one step. Then we can find the probability of arriving at any of these nodes in the previous step. In a specific row i of h_{K-1} , then, we have the sum of the probabilities that we start at node i and find a path through to S . This is exactly Ph_{K-1} . **I don't think this is very helpful.** We generate and store the vectors h_k for each step.

We also store the marginal probabilities. We use this for the entropy calculation and the probability mass calculation

To sample a path conditioned to start at node B and end at node S , we use a combination of our transition matrix and our vectors h_k . Firstly, we know that $X_0 = B$, so we can find

$$\Pr(X_1 = j | X_0 = B, X_k = s) = P_{ij} \frac{h_1(j)}{h_0(B)},$$

vectorised we have

$$Q_1(B, :) := P_{B, \cdot} \frac{h_1(\cdot)}{h_0(B)}.$$

Then we sample

$$X_1 \sim Q(B, :),$$

giving us the next point in the Markov bridge. We repeat the process finding

$$Q_2(X_1, :) = P_{X_1, \cdot} \frac{h_2(\cdot)}{h_1(X_1)},$$

and sampling

$$X_2 \sim Q_2(X_1, :).$$

We continue sampling until we reach X_K , which, by construction, is always our stressed endpoint. Repeating this process 10,000 times gives a distribution of paths from our benign to stressed point through the learned geometry.

To create our density-aware transition matrix, we simply create a column vector whose elements are $L = \exp(-\alpha V_j)$, and we multiply every column in P by the corresponding element in L . We then normalise this matrix so that the each row sums up to one, thus defining a new

transition matrix $\bar{P}$, where $\bar{P}_{ij}$ is the probability of node i jumping to node j in the density-aware scheme.

The only notion of the diffusion coordinates these Markov bridges have is the fact that their transition probabilities are varied by the density of the diffusion coordinates. However, the diffusion coordinates represent eigenfunctions of this diffusion process **Do they?**. We examine the structure of these graphs by mapping them to the corresponding diffusion coordinates. This is so that we can compare them to our density-aware graph paths **And I think that it is quite natural because of how the diffusion coordinates are defined by this matrix anyway, so there is a strong connection.**

For each event, we compute the density-aware path, and then we compute the Markov bridges, where we set the number of jumps in each Markov bridge to be the same as the number of jumps in the density-aware path for that event. This allows us to directly compare each of the computed metrics.

We compute the same metrics as we did in Section 4.6, namely we calculate

$$M_{\min}(\gamma), \quad M_{\text{mean}}(\gamma), \quad L_{\text{diff}}(\gamma), \quad T_{\text{turn}}(\gamma),$$

for each generated path. This results in a range of values. From this range, we inspect the distributions and compare them to our density-aware graph path.

We also compute two distribution-specific metrics, which we cannot compute for the deterministic density-aware path. Firstly, we calculate the entropy of the bridge paths, using the marginals $\mu_k(i)$ where k is the step in the Markov bridge and i is the node index. The entropy is defined as

$$H_k = - \sum_{i=1}^N f(\mu_k(i)),$$

where

$$f(\mu_k(i)) = \begin{cases} \mu_k(i) \log \mu_k(i) & \text{if } \mu_k(i) \neq 0, \\ 0 & \text{if } \mu_k(i) = 0. \end{cases}$$

Citation? The marginals $\mu_k(i)$ tell us the probability of the random variable k steps along the Markov bridge being node i . The tighter the spread of the Markov bridges, the smaller the entropy will be. Since $0 \leq \mu_k(i) \leq 1 \forall k, i$, then $\log \mu_k(i) \leq 0$ and $H_k \geq 0$. We compare the entropy of the Markov bridges for $\alpha = 0$, and $\alpha > 0$.

Next, we calculate the expected distance of a Markov bridge path to the density-aware path via

$$d_k = \mathbb{E} \left[\|\Psi(X_k) - \Psi(\Gamma_k^\beta)\| \right],$$

however, because we have stored the marginals, we can calculate this as

$$d_k = \sum_{i=1}^N \mu_k(i) \|\psi_i - \psi_{\Gamma_k^\beta}\|.$$

Finally, we can use the Markov bridges to generate distributions of paths in macroeconomic space where each point in every path corresponds to a historically supported combination of variables. We plot a confidence interval of a selection of macroeconomic variables for each stress event.

7.3 Markov bridges results

We first read the results of the entropy of the Markov bridges over an increasing value of α . Increasing α decreases the probability of jumps to low-density nodes, and therefore, one would expect the variation of the distribution to decrease as α increases. In Figure 11 we see exactly that. In Figure 11 we can see that increasing α has the effect of making the distribution of the Markov bridge much tighter (less variance). This is exactly what we expected and it's exactly

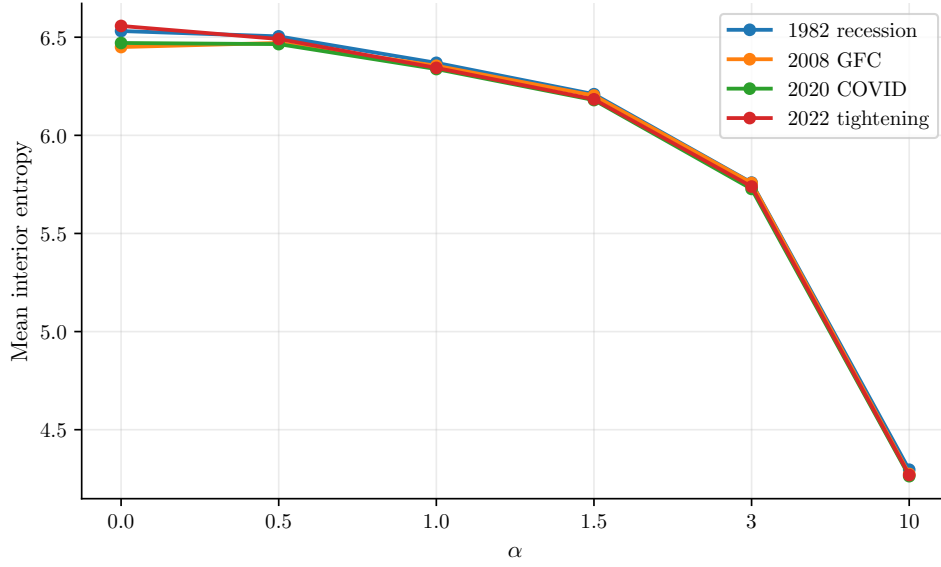


Figure 11: Caption

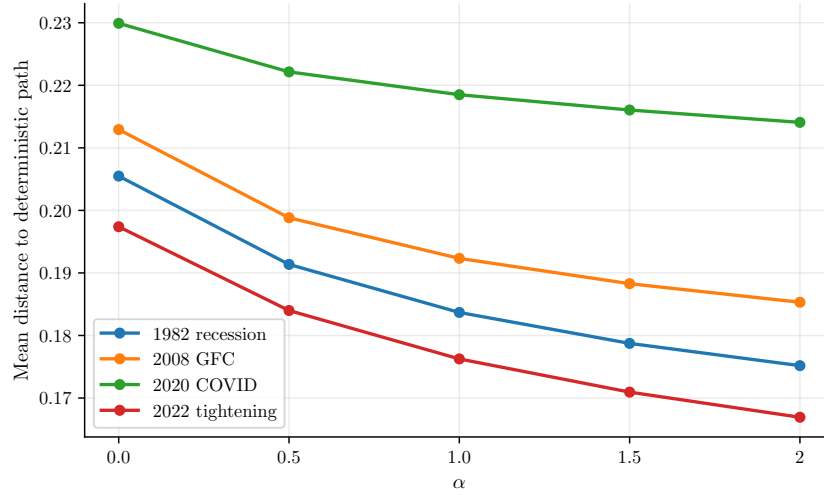


Figure 12: Caption

what we desired. A slight worry is that the entropy of each of the distributions looks very similar across events, *I am not sure if this is what you'd expect?*. This might suggest that regardless of the endpoints, the distribution of the interior of the Markov bridge is similar, which *I don't think this would be desired. Maybe this is how the economy works?*. I have checked and it doesn't look too bad. The means of the paths have different points; these points get closer together and further into the central high-density region as α increases, implying that increasing α so much means that the chain quickly forgets about the endpoints, it mainly just makes jumps between high density points and then makes another big jump to the endpoint when it needs to. This could be remedied by varying the density calculation to have a wider spread of high-density points, meaning that while avoiding sparse regions, the Markov bridge could still differentiate themselves by gradually changing distribution towards an endpoint, rather than all occupying the same dense region and then taking different large jumps to their respective stressed states. This may not be necessary if we observe that the economy does tend to bounce within a normal region and then jump to a stressed region. *Should I include a figure showing that the mean values of the points in diffusion space all start to go towards the central high density region? Maybe a density cap would be a good idea so that we have generally dense regions, where the path isn't directly forced right into the most dense part.*

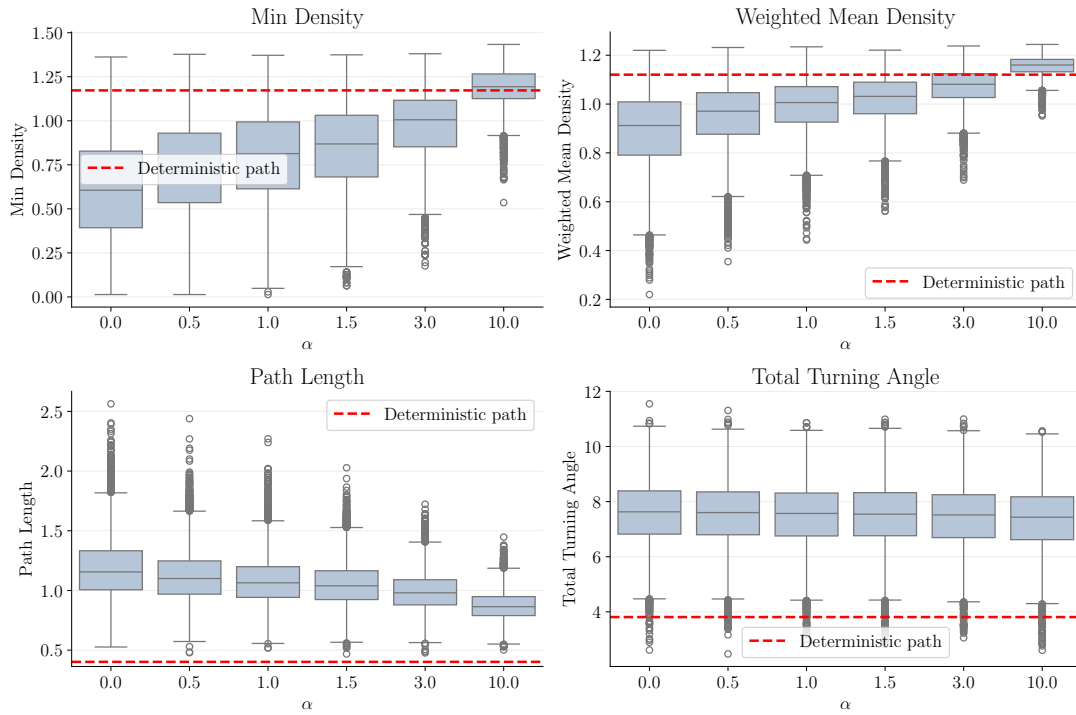


Figure 13: Caption

We now begin to compare the Markov bridges to the density-aware path in diffusion coordinates. In Figure 12 we can see that increasing α decreases the expected distance from the Markov bridge **I am not sure if I am using the correct terminology; is a Markov bridge a single path or is it the entire distribution?** to the density-aware path. This is an encouraging result as it suggests that our density-aware path may give us a meaningful path **Why? What would it mean if this was the case? Maybe discuss the limits as $\beta, \alpha \rightarrow \infty$. Maybe discuss how the means of the paths may converge towards each other in some conditions. It looks like.**

Now we discuss the average metrics of the graph paths displayed in Figure 13

- I think show box plots for one event and discuss them all
- then show the average results for all the events together.
- For some events, the min and mean density of the density-aware path is higher than most of the corresponding averages for the Markov bridges. Whereas for other events, the Markov bridge averages are higher than the density-aware metrics. This could reflect the different regions which they pass through. Maybe the dense regions are too far away for the density-aware paths (because of the distance penalty in the shortest path algorithm). What I am saying is that it might be a good idea to show all of the box plots for min and mean density metrics.
- Path length metrics are all very similar, just decreasing. I've not looked at turning angle metrics

Need to balance sparse region avoidance with short jumps

8 Hidden Regimes on the Macroeconomic Manifold

This is the extension

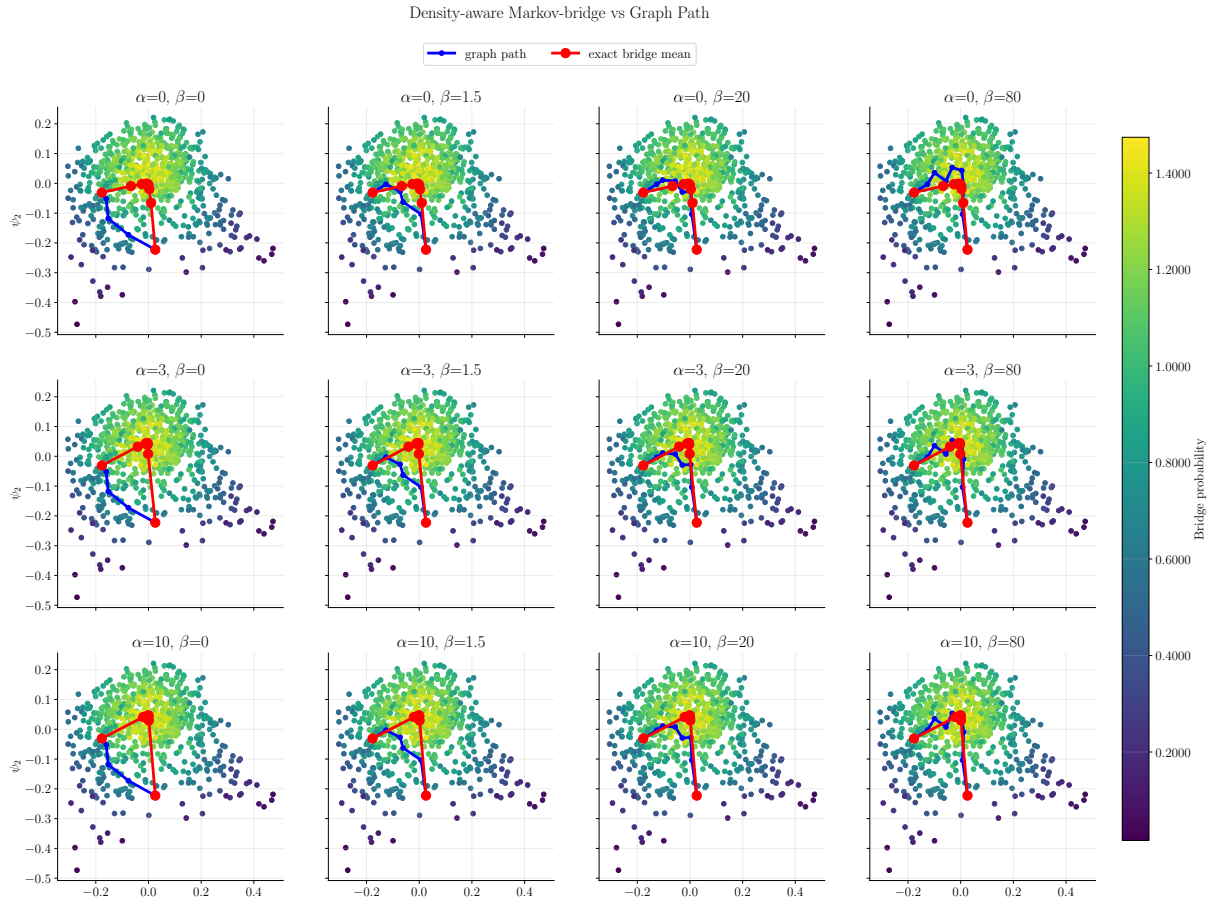


Figure 14: Caption

9 PCA benchmark results

10 Conclusions and outlook

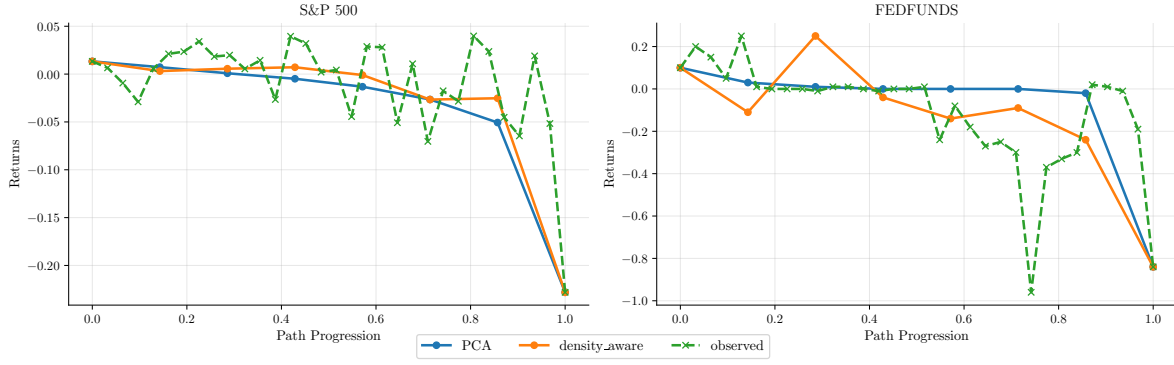
References

- [1] G. Baker, A. Capponi, and J. A. Sidaoui. Data-driven dynamic factor modeling via manifold learning. *arXiv preprint arXiv:2506.19945*, 2025.
- [2] C. Borio, M. Drehmann, and K. Tsatsaronis. Stress-testing macro stress testing: Does it live up to expectations? *Journal of Financial Stability*, 12:3–15, 2014. Reforming finance.
- [3] R. R. Coifman, I. G. Kevrekidis, S. Lafon, M. Maggioni, and B. Nadler. Diffusion maps, reduction coordinates, and low dimensional representation of stochastic systems. *Multiscale Modeling & Simulation*, 7(2):842–864, 2008.
- [4] R. R. Coifman and S. Lafon. Diffusion maps. *Applied and computational harmonic analysis*, 21(1):5–30, 2006.
- [5] A. W. Long and A. L. Ferguson. Landmark diffusion maps (l-dmaps): Accelerated manifold learning out-of-sample extension. *Applied and Computational Harmonic Analysis*, 47(1):190–211, July 2019.
- [6] D. Perraul-Joncas and M. Meila. Non-linear dimensionality reduction: Riemannian metric estimation and the problem of geometric discovery. *arXiv preprint arXiv:1305.7255*, 2013.
- [7] A. Singer, R. Erban, I. G. Kevrekidis, and R. R. Coifman. Detecting intrinsic slow variables in stochastic dynamical systems by anisotropic diffusion maps. *Proceedings of the National Academy of Sciences*, 106(38):16090–16095, 2009.

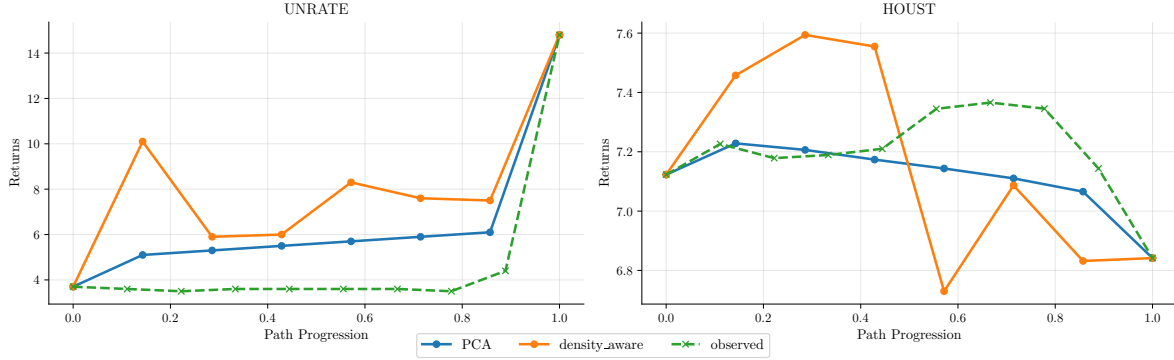
Appendices

A First appendix

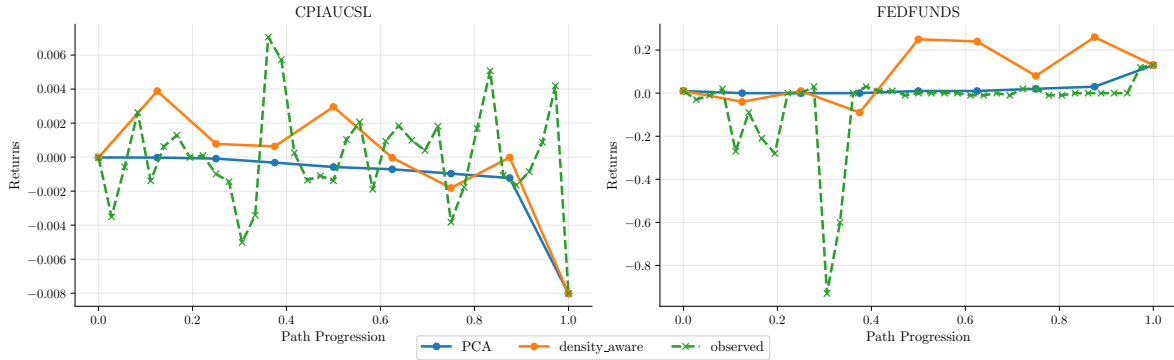
B Second appendix



(a) Transition from the benign state in 2006 to the Global Financial Crisis in 2008.



(b) Transition from the benign state in 2019 to the COVID-19 disruption in 2020.



(c) Transition from the benign state in 2019 to the monetary-tightening episode in 2022.

Figure 15: Lifted macroeconomic trajectories for the COVID-19 disruption and the 2022 monetary-tightening episode. The graph path consists of observed transformed macroeconomic states, whereas intermediate points on the PCA path are reconstructed pseudo-inverse of PCA map.